# Multi-Agent Uncertainty-Aware Pessimistic Model-Based Reinforcement Learning for Connected Autonomous Vehicles

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***Abstract*—Deep Reinforcement Learning (DRL) holds significant promise for achieving human-like Autonomous Vehicle (AV) capabilities, but suffers from low sample efficiency and challenges in reward design. Model-Based Reinforcement Learning (MBRL) offers improved sample efficiency and generalizability compared to Model-Free Reinforcement Learning (MFRL) in various multi-agent decision-making scenarios. Nevertheless, MBRL faces critical difficulties in estimating uncertainty during the model learning phase, thereby limiting its scalability and applicability in real-world scenarios. Additionally, most studies on Connected Autonomous Vehicles (CAVs) focus on single-agent decision-making. In contrast, existing multi-agent MBRL solutions lack computationally tractable algorithms with Probably Approximately Correct (PAC) guarantees, a crucial factor for ensuring policy reliability with limited training data. To address these challenges, we propose `MA-PMBRL`, a novel Multi-Agent Pessimistic Model-Based Reinforcement Learning framework for CAVs, incorporating a max-min optimization approach to enhance robustness and decision-making. To mitigate the inherent subjectivity of uncertainty estimation in MBRL and avoid incurring catastrophic failures in AV, `MA-PMBRL` employs a pessimistic optimization framework combined with Projected Gradient Descent (PGD) for both model and policy learning. `MA-PMBRL` also employs general function approximations under partial dataset coverage to enhance learning efficiency and system-level performance. By bounding the suboptimality of the resulting policy under mild theoretical assumptions, we successfully establish PAC guarantees for `MA-PMBRL`, demonstrating that the proposed framework represents a significant step toward scalable, efficient, and reliable multi-agent decision-making for CAVs.**

***Index Terms*—Autonomous vehicle control, multi-agent model-based reinforcement learning, probably approximately correct guarantee.**

## I. Introduction

RESEARCH on Connected Autonomous Vehicles (CAVs) has gained considerable momentum, particularly in developing advanced control algorithms [1], [2]. In mixed-traffic environments, CAVs integrate perception, planning, and control to cooperate with other vehicles and infrastructure, supporting tasks such as network-level traffic-flow optimization [3], tactical decision-making [4], and longitudinal/lateral control [5]. These demands have spurred increasing interest in RL-driven controllers, especially multi-agent methods [6]. However, partial observability, limited communication, nonstationary human behavior, and heterogeneous vehicle capabilities hinder practical deployment, motivating scalable, sample-efficient, and safety-aware multi-agent learning. MARL has emerged as a promising paradigm for autonomous CAV control. Traditionally, Model-Free RL (MFRL) has been widely adopted [7], [8], [9], yet its reliance on extensive interactions is costly, often leading to unstable training and high computational overhead in autonomous driving. Meanwhile, many vehicular-network algorithms follow the Centralized Training and Decentralized Execution (CTDE) framework [10], which can suffer from limited scalability and pronounced performance degradation under communication constraints or unreliability [11]. To address these issues, recent studies have explored Decentralized Training and Decentralized Execution (DTDE) algorithms [12] and adopted Model-Based RL (MBRL) [11], [13] for communication-constrained scenarios.

In contrast to MFRL, MBRL [11], [13] typically reduces real-world interactions by learning dynamics and reward models to support training, which is particularly appealing in CAV scenarios where data collection is costly or constrained and strong generalization under non-stationary conditions is required. By amortizing part of the interaction cost into computation via learned rollouts, MBRL achieves higher sample efficiency under a fixed interaction budget. However, model bias and error accumulation induced by *aleatoric* and *epistemic* uncertainty [14]

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can undermine policy stability and robustness, especially in planning schemes such as MPC [15], where multi-step predictions compound one-step errors [16]. This has motivated modified Dyna-style algorithms [17] to mitigate compounding errors. For example, MBPO [18] interleaves short-horizon model rollouts with off-policy updates on real data, improving sample efficiency while reducing model bias through conservative rollout horizons. Beyond compounding errors, MBRL faces a pronounced Out-Of-Distribution (OOD) issue when dataset coverage is limited, which hinders policy improvement. To address this, algorithms in the MBPO-style family, including MOPO [19], MOBILE [20], and MOReI [21], typically rely on idealized assumptions about uncertainty estimation. In practice, uncertainty quantification (e.g., truncation [13]) introduces subjectivity via heuristic thresholds and can compromise scalability and stability; moreover, explicit uncertainty estimates from deep neural networks are often unreliable [22]. These considerations motivate robust algorithms that avoid relying on explicit uncertainty estimation and its intrinsic limitations.

Robust MBRL [23], [24] aims to optimize worst-case performance over an uncertainty set of MDPs. By incorporating perturbations during training, it can improve stability and generalization in safety-critical CAV settings, sometimes at the cost of nominal performance. For instance, [22] proposes CPPO, a pessimistic offline MBRL method that searches for the worst-case transition model within a constrained MDP set and performs max–min policy optimization. Under general function approximation, CPPO establishes sub-optimality bounds and PAC guarantees under *partial coverage*, which relaxes the stringent requirement of bounded density-ratio concentrability for all learned policies by only assuming boundedness for the optimal policy [22]. Despite its theoretical value, CPPO is difficult to implement due to worst-case model search and max-min optimization. Building on CPPO, [25] develops RANMBO-RL, which improves robustness by generating pessimistic synthetic transitions for OOD data and reformulating policy optimization as a two-player zero-sum game against an adversarial environment model. In addition, [26] outlines a pessimistic framework, but lacks empirical validation and is limited to the single-agent setting. However, extending pessimistic MBRL to the multi-agent case with computationally tractable algorithms and PAC-style guarantees—crucial under partial-coverage datasets—remains largely unexplored. In our analysis, we explicitly adopt the standard *partial coverage* assumption (finite concentrability for the optimal policy), as defined in Definitions 1 and 2, and clarify its role in our guarantees.

In this paper, we investigate self-decision control for CAV-based traffic management under a DTDE framework, tailored to communication constraints and partially covered limited datasets. To provide a cost-effective MARL solution with theoretical guarantees, we propose MA-PMBRL, a decentralized pessimistic MBRL algorithm built on Soft Actor-Critic (SAC) [27]. MA-PMBRL enables efficient local sample sharing among neighboring vehicles within a communication range and incorporates a pessimistic model-based min-max framework that enhances safety without relying on subjective uncertainty estimation. By attenuating value estimates in low-coverage, high-volatility regimes (e.g., dense merges or lane changes near intersections), it induces appropriately conservative actions while preserving efficiency elsewhere. Moreover, unlike RANMBO-RL which formulates pessimism as a two-player zero-sum game, MA-PMBRL leverages Projected Gradient Descent (PGD) [28] within the critic to efficiently derive control policies from pessimistic estimates. The contributions of this work are summarized as follows:

- We propose a sample-efficient and computation-implementable Multi-Agent Pessimistic MBRL algorithm MA-PMBRL, built upon SAC and a max-min framework to enhance performance and robustness. MA-PMBRL effectively avoids subjective uncertainty estimation in MBRL without compromising performance while leveraging a fixed dataset with partial coverage to address covariate shifts in MBRL.
- Building upon the theoretical foundations of graph theory [29] and concentration inequalities [30], we provide a rigorous theoretical analysis of MA-PMBRL under communication constraints and partial coverage. Unlike [22] and [26], our work provides a more comprehensive comparative analysis for multi-agent settings. Specifically, we enhance and refine its theoretical derivations, addressing previously incomplete aspects of the original proofs. Additionally, our studies reveal how the communication range affects the convergence of MARL, which is further rigorously validated in challenging urban driving scenarios. Notably, unlike common MFRL methods, our result does not rely on the assumption of Bellman completeness [31].
- Within the multi-agent framework, we also propose a communication protocol that operates under communication constraints, thereby facilitating the efficient exchange and better balancing of communication overhead and system performance. Extensive simulations validate the superior sample efficiency of MA-PMBRL over other solutions like MA-PETS [32], SVMIX [9], FIRL [8], MBPO [18], SAC [27].

The remainder of the paper is organized as follows. Section III briefly introduces the mathematical preliminaries and formulates the problem to be investigated. In Sections IV and V, we provide more details of MA-PMBRL and establish an upper bound for the sub-optimality of the policy it yields, respectively. Finally, Section VI demonstrates the effectiveness and superiority of MA-PMBRL through extensive simulations. We conclude the paper in Section VII.

Beforehand, Table I summarizes mainly used notations.

## II. Related Works

### A. Multi-Agent Reinforcement Learning for CAVs

The complexities of urban high-density mixed-traffic intersections present both challenges and opportunities to improve traffic efficiency and safety through CAV interactions, with MARL garnering attention for its potential to enhance these aspects [6], [33]. Incorporating communication, such as Vehicle-to-Vehicle (V2V) interaction, improves coordination, reduces collisions, and enhances decision-making. In multi-agent CAV settings, two prevailing paradigms are CTDE and DTDE.

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TABLE I
THE KEY PARAMETERS USED IN THE PAPER

| Notation | Parameters Description |
|---|---|
| $I$ | Number of RL agents |
| $K$ | No. of episodes |
| $H$ | Length per episode |
| $\gamma$ | Discount factor |
| $\mathcal{M}_{\mathcal{D}}$ | Constrained MDP set driven by $\mathcal{D}$ |
| $\mathcal{D}_E$ | Buffer for storing actual data |
| $\mathcal{D}_{\tilde{T}_\phi}$ | Buffer for storing virtual data |
| $T$ | True dynamics model of the unknown environment |
| $T_\phi$ | Candidate model of constrained MDP set $\mathcal{M}_{\mathcal{D}}$ |
| $\tilde{T}_\phi$ | Solution of the pessimistic model-based min-max RL problem |
| $L_{\text{rollout}}$ | Rollout length |
| $X$ | Maximum iteration of PGD updates |
| $\phi_{\text{best}}$ | Smallest objective value of PGD |
| $d$ | Communication range |
| $\mathcal{B}_t^{(i)}$ | The set of neighboring CAVs for vehicle $i$ at $t$ |
| $v_t$ | Speed per vehicle at time-step $t$ |
| $z_t$ | Position per vehicle at $t$ |
| $\bar{v}_t$ | Target speed per vehicle at $t$ |
| $v_{t,a}, v_{t,e}$ | Speed of vehicles ahead and behind |
| $l_{t,a}, l_{t,e}$ | Distance of vehicles ahead and behind |
| $b^{(i)}$ | Vehicle in emergency braking range (1: yes, 0: no) |
| $\Lambda x_t$ | Travel distance per vehicle at $t$ |
| $C_{\pi^*}$ | Concentrability coefficient for $\pi^*$ |
| $\Delta_a$ | Performance gap between the true model and estimated model under $\pi^*$ |
| $\Delta_b$ | Gap between the pessimistic values of $\pi^*$ and $\pi_t$ over $\mathcal{M}_{\mathcal{D}}$ |
| $\Delta_c$ | Pessimism gap between the worst-case value over $\mathcal{M}_{\mathcal{D}}$ and the true-model value under $\pi_t$ |
| $\Gamma_a$ | Expected advantage function difference over $\mathcal{M}_{\mathcal{D}}$ |
| $\Gamma_b$ | Expectation of the weighted gradient of the policy log-likelihood over $\mathcal{M}_{\mathcal{D}}$ |
| $\Gamma_c$ | Difference in expected policy gradient terms over $\mathcal{M}_{\mathcal{D}}$ |
| $c_1$ | Scaling constant associated with approximation and concentration bounds |
| $c_2$ | Adjustment factor for complexity term, influencing worst-case error estimation at confidence level $1-\delta$ |
| $c_3$ | Upper bound on the maximum weighted feature $w$ and influence of feature $\varphi$ |
| $\mathbf{C}_d$ | Cliques cover of $\mathcal{G}_d$ |
| $\bar{\chi}(\mathcal{G}_d)$ | Minimum number of cliques cover |

*1) CTDE-Based Methods:* CTDE methods enable coordinated learning by leveraging shared or privileged information during training, while preserving decentralized decision-making at deployment. Within actor–critic MFRL, [34] introduces `MA-DDPG`, extending `DDPG` to multi-agent settings under the CTDE paradigm. Additionally, [35] proposes `COMA`, which improves credit assignment through counterfactual experience replay, enhancing multi-agent collaboration. However, as the number of agents or the action space grows, the computational burden of MFRL can scale combinatorially, creating a major bottleneck. To mitigate this, [9] proposes using a stochastic graph neural network to capture dynamic topological features of time-varying graphs, while decomposing the value function. To reduce communication overhead resulting from frequent information exchange, [8] introduces a consensus-based optimization approach built on periodic averaging, which integrates the consensus algorithm into Federated Learning (FL) for local gradient exchange. Recent works further provide principled privacy-preserving designs for FL with local gradient exchange, offering adaptive defenses against gradient leakage while maintaining learning utility [36], [37].

*2) DTDE-Based Methods:* Unlike CTDE, the DTDE paradigm eliminates reliance on a centralized training process, enabling agents to both train and act locally. This improves scalability and reduces coupling among agents by avoiding the exchange of local observations, gradients, or policy parameters, which in CTDE may raise privacy and security concerns, such as vehicle locations or trajectories [38]. Recent studies have adopted DTDE in CAV scenarios to improve robustness under non-stationarity and tight communication budgets, with applications such as cooperative navigation [39]. To stabilize independent agents without centralized coordination, [13] introduces an uncertainty-aware model-based RL method (`UA-MBRL`) within a DTDE framework that reduces both communication and computation overhead while maintaining competitive performance.

Despite the promising results of both paradigms, MFRL algorithms remain hindered by high computational and sampling demands, which limit their applicability in real-time CAV scenarios [40]. In particular, when data acquisition is costly and agent-environment interactions are constrained, performance can rapidly deteriorate and become unstable.

### B. Model-Based Reinforcement Learning

MBRL naturally emerges as an alternative solution to address the sampling efficiency and communication overhead issues in MFRL. Unfortunately, MBRL suffers from a performance deficiency, as it may fail to accurately estimate environmental uncertainty and characterize the dynamics model, a critical research component in MBRL. For example, PILCO [41] marginalizes the aleatoric and epistemic uncertainty of a learned dynamics model to optimize the expected performance. Most MBRL algorithms suffer from subjectivity in uncertainty estimation. Taking the example of the variants of `MBPO` [18], [19] introduces `MOPO`, which optimizes the policy under the uncertainty-penalized MDP framework. By incorporating a soft reward penalty based on model error estimates, the policy can return to high-confidence regions after taking risky actions, without being terminated. Meanwhile, [21] introduces `MoRel`, which defines terminal states by combining a subjective threshold based on uncertainty and uses a large negative reward to penalize unknown regions, thereby enhancing the robustness of decision-making in RL.

To address subjective uncertainty estimation and OOD issues, [22] proposes `CPPO`, which introduces a constrained pessimistic policy optimization to limit excessive optimization and improve stability in uncertain environments from offline data with partial coverage. Building on this, [25] introduces `RAMBO-RL`, which incorporates a two-player zero-sum game formulation against an adversarial environment model. The model is trained to minimize the value function while ensuring accurate predictions of the transition.

### C. Multi-Agent Model-Based Reinforcement Learning

Given the sample efficiency, faster learning, and convergence benefits of MBRL, there is growing interest in extending MBRL

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to multi-agent settings. For example, [7] introduces MAPPO, which integrates a prior model into the PPO framework to accelerate the learning process using current centralized coordination methods. Nonetheless, even with local models, many methods still rely on a global centralized critic. Such centralized coordination methods induce substantial computation and communication overheads, reduce flexibility, and introduce latency due to the global aggregation of system-wide information. Moreover, limitations would be exacerbated under bandwidth constraints [42]. Additionally, CTDE relies on global information during training, which poses challenges for implementation in real-world applications and leads to scalability issues in large-scale or resource-constrained deployments.

Several recent studies explore decentralized MBRL for limited communication vehicular networks. For example, Ref. [13] proposes an uncertainty-aware approach that leverages model ensembles and adaptive truncation to mitigate compounding error. Ref. [32] adopts probabilistic ensembling with trajectory sampling [32], while [11] proposes decentralized policy optimization with agent-level topological decoupling. Nevertheless, there remains limited work that explicitly addresses subjective uncertainty estimation in multi-agent MBRL [43]. Motivated by the substantial communication overhead of centralized coordination, we focus on a decentralized multi-agent MBRL design tailored to limited-communication CAV networks, emphasizing robustness, sample efficiency, and scalability.

## III. Preliminaries and System Model

This section provides a brief introduction to the fundamentals and necessary assumptions of the underlying MDPs, as well as a framework for single-agent pessimistic MBRL. Regarding the decision-making issue for CAVs, we highlight how to formulate the problem under the multi-agent case.

### A. Preliminaries

*1) Parallel Markov Decision Process:* The decision-making problem of CAVs can be formulated as a collection of *parallel time-homogeneous stochastic MDPs* $\mathcal{M} := \{M(\mathcal{S}^{(i)}, \mathcal{A}^{(i)}, \mathcal{P}^{(i)}, \mathcal{R}^{(i)}, \mu_0, \gamma, H)\}_{i=1}^{I}$ among agents $i \in [I] := \{1, 2, \ldots, I\}$ for $H$-length episodes [44]. Despite its restrictiveness, parallel MDPs provide a valuable baseline for generalizing to more complex environments, such as heterogeneous MDPs. Notably, agents have access to identical state and action space (i.e., $\mathcal{S}^{(i)} = \mathcal{S}^{(j)}$ and $\mathcal{A}^{(i)} = \mathcal{A}^{(j)}, \forall i, j \in [I]$), $\mathcal{R}^{(i)} : \mathcal{S}^{(i)} \times \mathcal{A}^{(i)} \to [-R_{\max}, R_{\max}]$ is the reward function, $\mathcal{P}^{(i)} : \mathcal{S}^{(i)} \times \mathcal{A}^{(i)} \times \mathcal{S}^{(i)} \to \mathbb{R}^+$ is the transition functions, $\mu_0$ is the initial state distribution, and $\gamma$ the discount factor. Furthermore, we focus exclusively on stationary policies, denoted as $\pi^{(i)} : \mathcal{S}^{(i)} \to \mathcal{A}^{(i)}$, which represents a decision strategy to pick an action $a_t^{(i)} \in \mathcal{A}^{(i)}$ with probability $\pi^{(i)}(\cdot|s, a)$ given the current state $s_t^{(i)} \in \mathcal{S}^{(i)}$.

In an MDP $M^{(i)}$, the standard objective is to identify a policy $\pi^{(i)}$ that maximizes the expected cumulative discounted reward, denoted as $V_{\mathcal{P},\pi}^{(i)}$, which is computed by integrating over the state space $\mathcal{S}^{(i)}$ under the transition dynamics $\mathcal{P}^{(i)}$, with an initial state distribution $\mu_0$ and a reward function $\mathcal{R}^{(i)}$, and is formally defined as:

$$V_{\mathcal{P},\pi}^{(i)} := \mathbb{E}_{\mathcal{P},\pi,s_0\sim\mu_0}\left[\sum_{t=0}^{\infty}\gamma^t\mathcal{R}^{(i)}\left(s_t^{(i)}, a_t^{(i)}\right) \mid s_0^{(i)}\right],$$

where $\gamma$ is the discount factor, and $s_t^{(i)}$ and $a_t^{(i)}$ are the state and action at time step $t$, respectively.

In parallel MDPs, the state-action value function, $Q_{\mathcal{P},\pi}^{(i)}$, captures the expected cumulative discounted reward when starting from state $s_0^{(i)}$ and taking action $a_0^{(i)}$ at the first time step, followed by a policy $\pi^{(i)}$ thereafter, defined as:

$$Q_{\mathcal{P},\pi}^{(i)} := \mathbb{E}_{\mathcal{P},\pi,s_0\sim\mu_0}\left[\sum_{t=0}^{\infty}\gamma^t\mathcal{R}^{(i)}\left(s_t^{(i)}, a_t^{(i)}\right) \mid s_0^{(i)}, a_0^{(i)}\right].$$

*2) Pessimistic Model-Based Min-Max Reinforcement Learning:* Common MBRL first learns an MDP model to facilitate policy training, since both transition and reward functions are unknown. After each episode, a dataset $\mathcal{D} := \{(s_t, a_t, r_t, s_t')\}_{t=1}^{n}$ is used to fit the dynamics and reward model (e.g., via MLE). For brevity, we use $T$ to denote the transition-reward kernel, where $T(s', r \mid s, a)$ is the probability of receiving reward $r$ and transitioning to $s'$ after taking action $a$ in state $s$. Accordingly, given the true MDP $M = (\mathcal{S}, \mathcal{A}, T, \mu_0, \gamma, H)$, the learned MDP is $\tilde{M} = (\mathcal{S}, \mathcal{A}, \tilde{T}, \mu_0, \gamma, H)$, which shares the same state/action spaces but replaces $T$ with the learned $\tilde{T}$. We further assume $\tilde{T}(\cdot \mid s, a)$ is parameterized by $\phi$.

Afterward, the goal of MBRL is to find the best possible policy contingent on the model learning from a fixed dataset, that is, $\tilde{\pi} = \arg\max_{\pi\in\Pi} V_{\tilde{M}}$. To minimize reliance on assumptions about uncertainty estimation, we try to find the policy $\tilde{\pi}$ by solving a constrained max-min optimization formulation [22] as

$$\tilde{\pi} = \arg\max_{\pi\in\Pi}\min_{T_\phi\in\mathcal{M}_\mathcal{D}} V_{T_\phi,\pi,\mu_0}, \tag{1}$$

subject to the fixed dataset $\mathcal{D}$-driven constrained MDP set of $\mathcal{M}_\mathcal{D}$

$$\mathcal{M}_\mathcal{D} = \left\{T_\phi \middle| \mathbb{E}_\mathcal{D}\left[\mathrm{TV}\left(\tilde{T}_{\mathrm{MLE}}(\cdot \mid s, a), T_\phi(\cdot \mid s, a)\right)^2\right] \le \xi\right\}, \tag{2}$$

where $\mathrm{TV}(P_1, P_2) = \frac{1}{2}\sum_{(s,a,s')\in\mathcal{D}}\left|P_1(s' \mid s, a) - P_2(s' \mid s, a)\right|$ denotes the total variation distance [45] between $P_1$ and $P_2$, $\tilde{T}_{\mathrm{MLE}}$ is the MLE dynamics model fitted on $\mathcal{D}$, and $\xi > 0$ specifies the allowable deviation. The resulting constrained MDP set $\mathcal{M}_\mathcal{D}$ in (2) mitigates large statistical errors under limited dataset coverage without requiring explicit uncertainty estimation. By allowing a controlled deviation from $\tilde{T}_{\mathrm{MLE}}$, the radius $\xi$ trades off robustness and adaptability, improving generalization to unseen state-action pairs while remaining consistent with observed data.

*3) Theoretical Motivation:* In traditional RL theory, let $d_{T,t}^{\pi} \in \Delta(\mathcal{S}\times\mathcal{A})$ denote the distribution of $(s_t, a_t)$ at time $t$ induced by policy $\pi$ under dynamics $T$. The discounted occupancy measure is $d_T^{\pi}(s, a) := (1-\gamma)\sum_{t=0}^{\infty}\gamma^t d_{T,t}^{\pi}(s, a)$. A common *global coverage* assumption requires the density-ratio concentrability $d_T^{\pi}(s, a)/\rho(s, a) \le C$ for all $\pi \in \Pi$, where $\rho$

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is the state-action distribution from which $\mathcal{D}$ is sampled. This assumption is often unrealistic in large-scale problems such as CAV control, where obtaining sufficiently exploratory data is difficult. Hence, we do not assume global coverage; instead, we adopt a *partial coverage* setting in which $d_T^{\pi^*}(s,a)/\rho(s,a)$ is bounded only for the optimal policy $\pi^*$.

*Definition 1 (Partial Coverage from [22]):* Partial coverage [46] holds if there exists an optimal policy $\pi^*$ such that:

$$\sup_{(s,a)} \frac{d_T^{\pi^*}(s,a)}{\rho(s,a)} \leq C,$$

where $C$ is a constant.

*Remark:* In the context of CAVs, partial coverage implies that the available driving dataset covers only a subset of commonly observed traffic states and interactions, while rare or safety-critical CAV scenarios are sparsely represented. This partial coverage assumption is significantly weaker than global coverage and more practical. It enables us to compete against the optimal policy $\pi^*$ that is sufficiently represented in a fixed dataset and achieve near-optimal performance under more realistic assumptions for MBRL scenarios. Notably, the concentrability coefficient for a policy $\pi$, which is generally adopted for controlling the distribution shift between the fixed dataset distribution and the occupancy measure induced by $\pi$, is computationally intractable. Therefore, we introduce the following new concentrability coefficient based on the model class $\mathcal{M}$ to facilitate computation.

*Definition 2 (Concentrability coefficient from [22]):* For an optimal policy $\pi^*$, we define the concentrability coefficient $C_{\pi^*}$ as:

$$C_{\pi^*} = \max_{T_\phi \in \mathcal{M}_\mathcal{D}} \frac{\mathbb{E}_{(s,a)\sim d_T^{\pi^*}}\left[\mathrm{TV}\left(T_\phi(\cdot \mid s,a), T(\cdot \mid s,a)\right)^2\right]}{\mathbb{E}_{(s,a)\sim \rho}\left[\mathrm{TV}\left(T_\phi(\cdot \mid s,a), T(\cdot \mid s,a)\right)^2\right]}.$$

The theoretical analysis from [22] shows that solving a constrained max-min optimization formulation, as shown in (1) outputs a policy, which, with high probability, is approximately as good as any policy with a state-action distribution covered by the dataset. This is formally stated in the following lemma.

*Lemma 1 ((9) from [22]):* Let $T$ denote the true MDP transition function, and define $\pi^*$ as the optimal policy. Meanwhile, let $\tilde{T}_\phi$ be the solution MDP obtained from solving the pessimistic model-based min-max RL problem for the dataset $\mathcal{D}$. Then, with probability at least $1-\delta$, for any MDP $T_\phi \in \mathcal{M}_\mathcal{D}$, the following inequality holds:

$$\begin{aligned} V_{T,\pi^*} - V_{\tilde{T}_\phi,\pi^*} &\leq c_1(1-\gamma)^{-2}\sqrt{\Lambda_{(s,a)\sim d_T^{\pi^*},\max}} \\ &\leq c_1\left(1-\gamma\right)^{-2}\sqrt{C_{\pi^*}\Lambda_{(s,a)\sim\rho,\max}}, \end{aligned}$$

where

$$\Lambda_{(s,a)\sim\rho,\max} = \sup_{T_\phi\in\mathcal{M}_\mathcal{D}} \mathbb{E}_{(s,a)\sim\rho}\ \mathrm{TV}\left(T_\phi(\cdot \mid s,a), T(\cdot \mid s,a)\right)^2.$$

Here, $C_{\pi^*}$ is defined in Definition 2, $\rho$ represents the state-action distribution from which $\mathcal{D}$ is sampled, and $c_1$ is a scaling constant associated with approximation and concentration bounds.

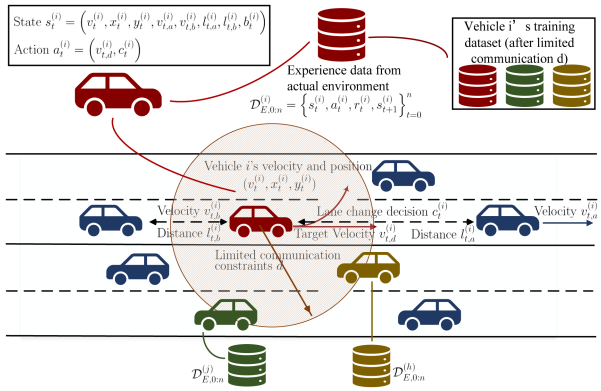



Fig. 1. System modeling diagram illustrating vehicle-state-action, local communication, and data availability.

Lemma 1 implies that solving (1) over the constrained MDP set in (2) yields a pessimistic model $\tilde{T}_\phi$ whose induced performance gap relative to the true dynamics $T$ under the optimal policy $\pi^*$ covered by the dataset is provably bounded.

### B. System Model

We consider a mixed-autonomy urban traffic system with $I$ CAVs, Human-driven Vehicles (HVs), and multiple intersections, as illustrated in Fig. 1. For each vehicle $i$, following the parallel MDP framework, the state $s_t^{(i)} \in \mathcal{S}^{(i)}$ comprises its speed $v_t^{(i)}$, position $z_t^{(i)} = (x_t^{(i)}, y_t^{(i)})$, the speeds and distances of the nearest vehicles ahead and behind $(v_{t,a}^{(i)}, v_{t,b}^{(i)}, l_{t,a}^{(i)}, l_{t,b}^{(i)})$, and a binary indicator $b_t^{(i)}$ denoting whether a vehicle lies within the emergency braking range. Each vehicle is controlled by a target speed $v_{t,d}^{(i)}$ and a binary lane-change decision $c_t^{(i)}$[1], yielding the action $a_t^{(i)} = (v_{t,d}^{(i)}, c_t^{(i)})$. We assume that both transition dynamics and reward functions are unknown and learned via neural networks through interactions with the environment.

Within the framework of decentralized multi-agent MBRL, we assume the availability of history state transition dataset for each agent, denoted as $\mathcal{D}_{E,0:n}^{(i)} = \{s_t^{(i)}, a_t^{(i)}, r_t^{(i)}, s_{t+1}^{(i)}\}_{t=0}^{n}$. Each agent approximates its transition model $\tilde{T}_\phi^{(i)}, \forall i \in [I]$ using DNNs parameterized by $\phi$, based on its own dataset $\mathcal{D}_{E,0:n}^{(i)}$ and exchanged samples from neighboring agents. In particular, within a graph $\mathcal{G}$ consisting of $I$ CAVs, each CAV can exchange its latest $H$-length[2] dataset $\{(s_t^{(i)}, a_t^{(i)}, r_t^{(i)}, s_{t+1}^{(i)})\}_{t_k-H}^{t_k}$ with neighboring CAVs within a communication range $d \in [0, D(\mathcal{G})]$ at the end of an episode $k$, where $D(\mathcal{G})$ represents the diameter of the graph $\mathcal{G}$. When $d=0$, the multi-agent MBRL problem reduces to $I$ independent single-agent cases. Mathematically, the dataset used for dynamics modeling at each CAV $i$ is updated as:

$$\begin{aligned} \mathcal{D}_{E,0:t_k}^{(i)} := \mathcal{D}_{E,0:t_k-H-1}^{(i)} \cup \{s_t^{(i)}, a_t^{(i)}, r_t^{(i)}, s_{t+1}^{(i)}\}_{t_k-H}^{t_k} \\ \cup_{j\in\mathcal{B}_{t_k}^{(i)}} \{(s_t^{(j)}, a_t^{(j)}, r_t^{(j)}, s_{t+1}^{(j)})\}_{t_k-H}^{t_k}, \end{aligned} \tag{3}$$

[1]In single-lane settings, $c_t^{(i)} \equiv 0$, and the action reduces to speed control.

[2]Notably, the applied length could be episode-dependent but is limited by the maximum value $H$, as will be discussed in Section IV.

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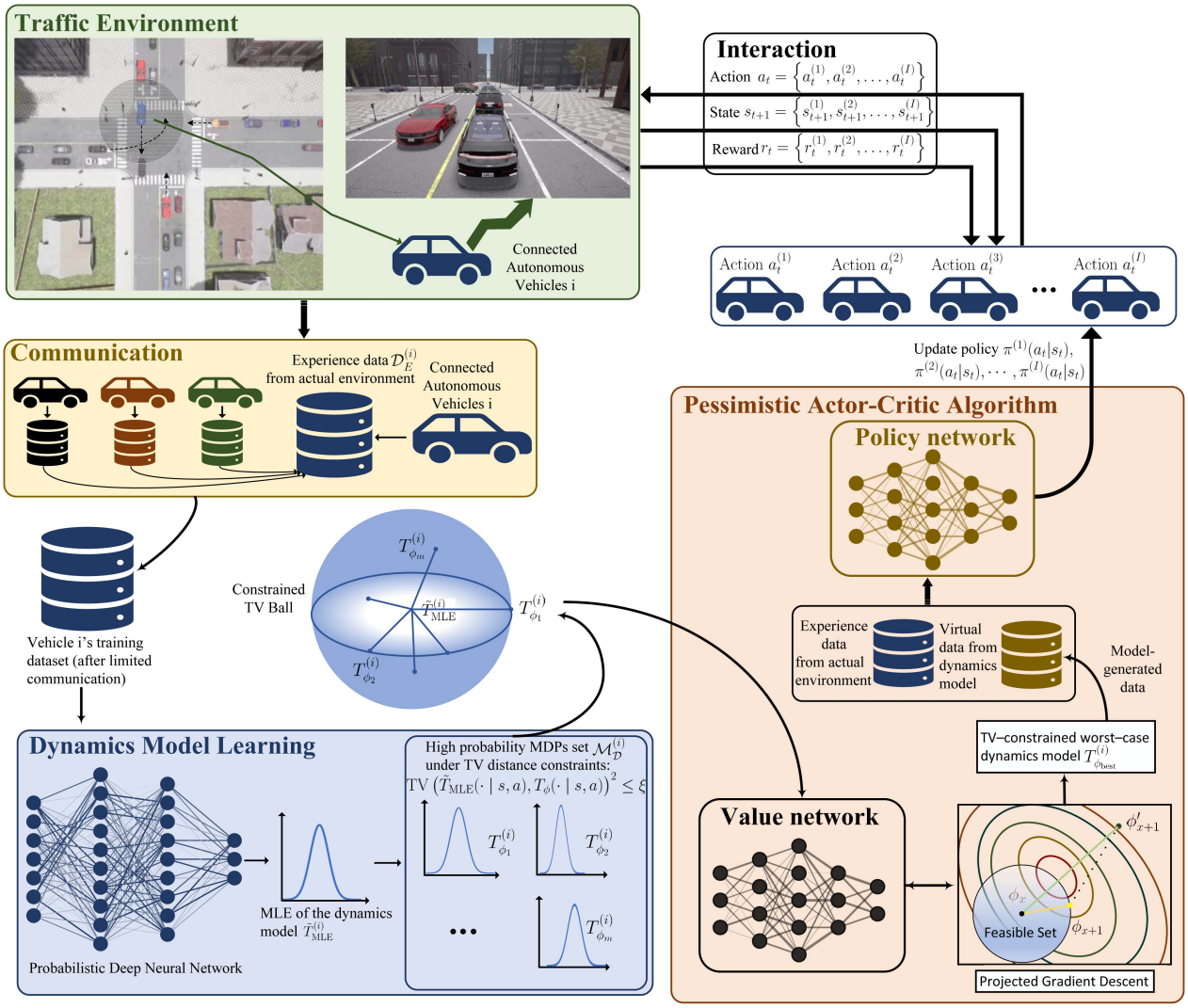



Fig. 2. The illustration of the MA-PMBRL algorithm for CAVs.

where $j \in \mathcal{B}_{t_k}^{(i)}$ represents neighboring CAVs that satisfy the Euclidean distance $\text{dis}_{t_k}(i,j) \le d$ for between CAVs $i$ and $j$ at the last time-step $t_k$ of episode $k$. Exchanging raw experience trades higher per-round communication cost for improved sample efficiency and stability under heterogeneous and safety-critical traffic conditions, which is the primary focus of this work.

The aforementioned multi-agent settings in a communication range $d$-limited environment transform the pessimistic problem in (1) as:

$$\tilde{\pi}^{(i)} = \arg\max_{\pi\in\Pi} \min_{T_\phi \in \mathcal{M}_{\mathcal{D}}} \mathbb{E}_{T_\phi,\pi,s_0\sim\mu_0}\left[\sum_{t=0}^{\infty}\gamma^t r_t^{(i)} \mid s_0^{(i)}\right],$$
$$s.t.\quad T_\phi^{(i)} \propto \mathcal{D}_{E,0:t}^{(i)}, 0 \le d \le D(\mathcal{G});$$
$$s_0^{(i)} \sim \mu_0; s_{t+1}^{(i)}, r_t^{(i)} \sim T_\phi^{(i)}(s_t^{(i)}, a_t^{(i)}), \forall t \in [H], \qquad (4)$$

where $\mathbb{E}(\cdot)$ denotes the expectation operator. This paper aims to provide a modal-based computation-tractable implementation to compute (4) and analyze its theoretical bound.

## IV. The MA-PMBRL Algorithm

In this section, we introduce a multi-agent MBRL framework that incorporates the pessimism principle into SAC, tailored for CAV scenarios, where both the dynamics model learning and policy decision-making processes in MA-PMBRL are illustrated in Fig. 2.

### A. Virtual Environment Dynamics Model Learning

Within the developed RL framework, an environment model is first constructed to enable rapid, uncertainty-aware prediction of interactions between the virtual agent and the environment. Without loss of generality, given available dataset $\mathcal{D}_{E,0:n}^{(i)} = \{(s_t^{(i)}, a_t^{(i)}, s_{t+1}^{(i)}, r_t^{(i)})\}_{t=0}^n$, we approximate the dynamics model at time-step $t$ (i.e., $\tilde{T}_\phi(s_{t+1}^{(i)}, r_t^{(i)} \mid s_t^{(i)}, a_t^{(i)}), \forall i \in [I]$), using a probabilistic DNN. Notably, assume that the conditional probability distribution of $s_{t+1}^{(i)}$ and $r_t^{(i)}$ follows a Gaussian distribution, the Maximum Likelihood Estimate (MLE) $\tilde{T}_{\text{MLE}}^{(i)}(\cdot \mid s_t, a_t) = \mathcal{N}(\tilde{\mu}_\phi(s_t,a_t), \tilde{\sigma}_\phi^2(s_t,a_t))$ of the transition model from

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the dataset MLE can be attained as:

$$\tilde{T}^{(i)}_{\text{MLE}} = \arg\min_{(s,a,s',r)\sim\mathcal{D}^{(i)}_{E,0:n}} \times \sum_{t=1}^{n}\left(\frac{\left(e^{(i)}_t-\tilde{\mu}_\phi\right)^2}{2\tilde{\sigma}^2_\phi}+\frac{\ln(2\pi\tilde{\sigma}^2_\phi)}{2}\right), \quad (5)$$

where $e^{(i)}_t$ denote the concatenation of the next step's state and reward $(s^{(i)\prime}, r^{(i)})$, and $\tilde{\mu}_\phi, \tilde{\sigma}^2_\phi$ denote the predicted mean and variance, thereby fitting the full conditional distribution—capturing the data's first- and second-order moments—and explicitly modeling aleatoric noise.

Next, we compute a constrained MDP set $\mathcal{M}^{(i)}_{\mathcal{D}_E}$, as defined in (2). While we adopt the TV distance for the theoretical modeling and analysis of worst-case distributional shift, the TV distance is less practical during implementations due to the cumbersome computations of an $L_1$ distance between high-dimensional continuous distributions. Therefore, we resort to the Kullback-Leibler (KL) divergence, an upper bound of the TV distance [47], as it provides a more tractable alternative for measuring the difference between two probability distributions. Thus, (2) will be reformulated as:

$$\mathcal{M}^{(i)}_{\mathcal{D}_E} = \left\{T^{(i)}_\phi \mid \mathbb{E}_{(s,a)\sim\mathcal{D}^{(i)}_E}\left[D_{\text{KL}}\left(T^{(i)}_\phi \| \tilde{T}^{(i)}_{\text{MLE}}\right)\right] \le \xi\right\} \quad (6)$$

Notably, the KL divergence between a Gaussian transition distribution $T^{(i)}_\phi(\cdot \mid s,a) = \mathcal{N}(\mu_\phi(s,a), \sigma^2_\phi(s,a))$, where $\mu_\phi$ and $\sigma_\phi$ represent the mean and diagonal covariance of the Gaussian distribution, respectively, and the MLE estimate $\tilde{T}_{\text{MLE}}(\cdot \mid s,a)$ can be expressed as:

$$D_{\text{KL}}\left(T^{(i)}_\phi \| \tilde{T}^{(i)}_{\text{MLE}}\right) = \frac{\sigma^2_\phi + (\mu_\phi - \tilde{\mu}_\phi)^2}{2\tilde{\sigma}^2_\phi} + \log\frac{\tilde{\sigma}_\phi}{\sigma_\phi} - \frac{1}{2}.$$

*Remark:* The constrained MDP set $\mathcal{M}^{(i)}_{\mathcal{D}_E}$ in (6) bounds distributional shift relative to the data distribution $\rho$ and yields a high probability feasible set under partial coverage. Unlike subjective uncertainty estimates (e.g., truncation thresholds), such a measurable discrepancy aligns with our training objective and theoretical transfer, enabling principled worst-case model selection without hand-tuned uncertainty quantification metrics.

### B. Constrained Pessimistic Soft Actor-Critic Algorithm

Building on the virtual environment model described above, SAC-based RL is the foundational algorithm for the proposed MA-PMBRL, which enhances learning efficiency and asymptotic performance.

In MA-PMBRL, the critic provides a pessimistic evaluation of the value function by performing PGD over a constrained MDP set. Compared to subjective, hand-tuned uncertainty quantification metrics, as built upon [48], it enables the principled selection of a robust, always-operable model, thereby avoiding reliance on subjective uncertainty quantification and facilitating principled worst-case model selection. Concretely, at each iteration $t$, given the current policy $\pi^{(i)}_t$, the critic of each agent $i$ selects *worst-case* model $\tilde{T}^{(i)}_\phi = \arg\min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t}$ from the constrained MDP set $\mathcal{M}^{(i)}_{\mathcal{D}_E}$. This introduces pessimism by ensuring that, with high probability, $V^{(i)}_{\tilde{T}_\phi,\pi_t} \le V^{(i)}_{T,\pi_t}$, since $T \in \mathcal{M}_{\mathcal{D}^{(i)}_E}$ with high probability.

Methodologically, MA-PMBRL departs from [48], where the critic assumes a *linear* $Q$ function and obtains a lower bound by minimizing $Q$ over a constrained coefficient set, whereas we minimize a *nonlinear* $Q$ over an appropriately constrained MDP set via PGD [28]. Specifically, the PGD update of $\phi^{(i)}$ under $\tilde{T}^{(i)}_\phi$[3] After each episode $k$ is performed as follows:

$$\begin{aligned}\phi^{(i)}_{x+1} &= \text{Proj}_{\mathcal{M}^{(i)}_{\mathcal{D}_E}}\left(\phi^{(i)\prime}_{x+1}\right)\\ &= \text{Proj}_{\mathcal{M}^{(i)}_{\mathcal{D}_E}}\left(\phi^{(i)}_x - \alpha_{\text{PGD}}\nabla_\phi Q^{(i)}_{\tilde{T}_{\phi_x},\pi_t}\right),\end{aligned} \quad (7)$$

where $\text{Proj}_{\mathcal{M}^{(i)}_{\mathcal{D}_E}}(\phi^{(i)\prime}_{x+1}) = \arg\min_{T_\phi\in\mathcal{M}^{(i)}_{\mathcal{D}_E}} \|\phi^{(i)\prime}_{x+1} - \phi^{(i)}\|_2$ represents the projection of $\tilde{T}^{(i)}_{\phi'_{x+1}}$ onto the constrained MDP set $\mathcal{M}^{(i)}_{\mathcal{D}_E}$, ensuring that the updated solution $\phi^{(i)}_{x+1}$ lies within the feasible region defined by $\mathcal{M}^{(i)}_{\mathcal{D}_E}$. Our approach has multi-fold merits. First, it can avoid the limitations of Bellman-restricted closeness and extend to more general MDPs with nonlinear $Q$ functions [22]. Besides, the adoption of PGD, which combines gradient computation and projection to systematically optimize the pessimistic estimation objective while maintaining theoretical rigor and numerical stability, ensures rapid convergence to a feasible solution. In safety-critical CAV settings, this translates to robustness against sensing or communication noise, occlusions, and non-stationarity, as the critic explicitly plans against plausible worst-case dynamics consistent with the data.

After executing a predefined maximum number of PGD updates, denoted as $X$, the algorithm generates a candidate point $\phi^{(i)}_x$ at each iteration $(x = 1, 2, \ldots, X)$. Upon completion, the algorithm returns the candidate point that achieves the smallest objective function value[4]:

$$\phi^{(i)}_{\text{best}} = \arg\min_{x\in[X]} Q^{(i)}_{\tilde{T}_{\phi_x},\pi_t} \quad (8)$$

In SAC, the actor-network is typically optimized using the Adam optimizer, which minimizes the gradient of the policy loss $\mathcal{L}^{(i)}_\pi$ concerning its parameters $\psi$, and the optimization objective for the actor policy network is typically achieved by minimizing the following loss function:

$$\mathcal{L}^{(i)}_\pi = \mathbb{E}_{s_t\sim\mathcal{D}^{(i)}_E\cup\mathcal{D}^{(i)}_{\tilde{T}_\phi}, a_t\sim\pi^{(i)}_\psi}\Big[\alpha_{\text{temp}}\cdot\log\tilde{\pi}^{(i)}(a_t \mid s_t) - Q^{(i)}_{\tilde{T}_\phi,\tilde{\pi}_t}(a_t \mid s_t)\Big] \quad (9)$$

Where $\tilde{T}^{(i)}_\phi$ is a Gau ssian distribution derived from the reparameterization trick, enabling the policy network to remain

[3]For simplicity of representation, we slightly abuse the notation $\phi^{(i)}$ for $\tilde{T}^{(i)}_\phi$ and $T^{(i)}_\phi$ here.

[4]At the end of $X$ iterations, there are three ways to select the output: return the final iterate $\phi_X$, the average of all iterates $\phi_{\text{avg}}$, or the point with the smallest objective value $\phi_{\text{best}}$[28]. Simulation results in Section VI-B demonstrate that selecting the smallest value yields the best performance.

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**Algorithm 1:** The Training of MA−PMBRL.

**Initialize:** Communication range $d$, initial state $\mathcal{N}(\mu_{\phi_0}, \sigma^2_{\phi_0})$.;
**Initialize:** Vehicle's policy network $\pi^{(i)}_{\psi}(\cdot \mid s) = \text{Unif}(\mathcal{A}^{(i)})$ and value networks $Q^{(i)}_{\tilde{T}_\phi,\pi}$;
**Initialize:** Vehicle's experience buffer $\mathcal{D}^{(i)}_{E} = \emptyset, \forall i \in [I]$ for storing actual data and $\mathcal{D}^{(i)}_{\tilde{T}_\phi} = \emptyset, \forall i \in [I]$ for storing virtual data;
Randomly sample action in the initial episode and update $\mathcal{D}^{(i)}_{E}, \forall i \in [I]$;
**for** *Vehicle* $i$ *and episode* $k$, $\forall i \in [I], \forall k \in [K]$ **do**
  $\tilde{T}^{(i)}_{\text{MLE}} \leftarrow$ Train the virtual environment MLE dynamics model on $\mathcal{D}^{(i)}_{E}$ for each vehicle $i$;
  **for** *step* $t = 1, 2, \ldots, H$ **do**
    **Generate synthetic** $l$**-step rollouts:**
    Let $l = t + 1$;
    **while** $l \leq t + L_{rollout}$ **do**
      Execute $a^{(i)}_l = \pi^{(i)}_{\psi}(s^{(i)}_l)$ in $\tilde{T}^{(i)}_{\phi}$;
      Get $s^{(i)}_{l+1}, r^{(i)}_l$ and add transition data $(s^{(i)}_l, a^{(i)}_l, r^{(i)}_l, s^{(i)}_{l+1})$ to $\mathcal{D}^{(i)}_{\tilde{T}_\phi}$;
      Let $l \leftarrow l + 1$;
    **Agent update:**
    Select the parameter $\phi^{(i)}_{\text{best}}$ after performing $X$ iterations of PGD, according to Eq. (7) and Eq. (8);
    Update $Q^{(i)}_{\tilde{T}_\phi,\pi}(s,a)$ and $\pi^{(i)}$ using samples from $\mathcal{D}^{(i)}_{E} \cup \mathcal{D}^{(i)}_{\tilde{T}_\phi}$;
    Execute $a^{(i)}_t = \pi^{(i)}_{\psi}(s^{(i)}_t), \forall i \in [I]$;
    Obtain reward $r^{(i)}_t$, and observe next-step state $s^{(i)}_{t+1}$;
  Determine neighboring vehicle $j, \forall \text{dis}(i,j) \leq d$ (i.e., within the communication range $d$ of vehicle $i$) to exchange information;
  Update dataset $\mathcal{D}^{(i)}_{E}$ by using Eq. (3).

differentiable. The term $\tilde{\pi}^{(i)}(a_t \mid s_t)$ is the output of the policy network (actor), representing the probability of taking action $a_t$ in state $s_t$. $Q^{(i)}_{\tilde{T}_\phi,\tilde{\pi}_t}$ is the estimated action-value function (critic), and $\alpha_{\text{temp}}$ is the temperature parameter that balances the contribution of policy entropy and the $Q$-value. Notably, for error compounding control, the loss function in (9) is minimized while we interleave *short–horizon* model rollouts with real data. The combination of short horizons and the bounded OOD drift jointly improves stability under partial coverage.

Finally, we summarize our model-based MARL method MA−PMBRL in Algorithm 1.

## V. Theoretical Guarantees of MA−PMBRL

In this section, we first provide a multi-agent theoretical guarantee of MA−PMBRL, by establishing the suboptimality of the learned policy $\pi_t$ from Algorithm 1 relative to an ideal optimal policy $\pi^*$, which need not be directly implemented but serves as an upper performance bound or convergence assurance based on the fixed data. Afterward, we derive the group PAC guarantee enabled by communication within a distance of $d$.

### A. Methodology

Beforehand, we introduce the assumption essential for our theoretical analysis to illustrate sample efficiency better. In our multi-agent CAVs framework, data are simultaneously and independently collected by multiple vehicles, each following its strategy. Therefore, the following assumption of dataset independence regarding the collection of fixed data can be made.

*Assumption 1 (Collection of Fixed Dataset):* The dataset of each vehicle $i$, $\mathcal{D}^{(i)}_{E,0:n} = \{(s^{(i)}_t, a^{(i)}_t, r^{(i)}_t, s^{(i)}_{t+1}) : t = 0, \ldots, n\}$ satisfies the i.i.d. $(s^{(i)}_t, a^{(i)}_t) \sim \rho$ with $s^{(i)}_{t+1} \sim T(\cdot \mid s_t, a_t)$, where $T$ is the true dynamics function of the unknown environment and $\rho$ denotes the distribution induced by the behavior policy under $T$.

To facilitate a clear and tractable theoretical analysis of policy optimization, we introduce the following assumption regarding the policy class.

*Assumption 2 (Log-linear Parametric Policy Class):* The policy class considered in this work is a log-linear parametric family defined as:

$$\Pi_\psi = \left\{ \pi_\psi(a|s) = \frac{\exp(\psi \cdot \varphi_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\psi \cdot \varphi_{s,a'})} \mid \psi \in \mathbb{R}^d \right\},$$

where the characteristic vector $\varphi : S \times A \to \mathbb{R}^d$ satisfies $\|\varphi_{s,a}\|_2 \leq \varphi_{\max}$ for any $(s,a)$. This assumption enables tractable theoretical analysis of policy optimization performance. More general parametric policy classes may be adopted under additional assumptions.

To find a policy for which we can provide a non-asymptotic bound on its sub-optimality $V_{T,\pi^*} - \max_{0 \leq t \leq H} V_{T,\pi_t}$, at the end of each episode $k \in [K]$ based on the fixed dataset, we define the group bound as:

$$\Delta^k_G(H) := \sum_{i=1}^{I} \left[ V^{(i)}_{T,\pi^*} - \max_{0 \leq t \leq H} V^{(i)}_{T,\pi_t} \right]. \tag{10}$$

Next, based on the fact that the mean lower bounds maximum, we decompose the cumulative sub-optimality $\Delta^k_G(H)$ in (10) into the sum of three parts as (11), shown at the bottom of the next page. We first investigate each part at the level of an individual agent and then present the group PAC bound.

### B. PAC Bound of Each Agent

In this part, we analyze an *implementable* SAC-based gradient procedure, thereby complementing the conceptual framework in [22] with a policy-learning error bound. Crucially, [22] provides bounds only for these model errors (i.e., $\Delta^{(i)}_a$ and $\Delta^{(i)}_c$)[5] and does not analyze the policy-optimization error $\Delta^{(i)}_b$. In contrast, we derive an explicit bound for $\Delta^{(i)}_b$, which quantifies

[5]Prominently, Appendix B states that, consistent with [22], the model-mismatch terms satisfy $\Delta^{(i)}_a \leq c_1(1-\gamma)^{-2}\sqrt{\frac{C_{\pi^*}\ln(c_2|\mathcal{M}_\mathcal{D}|/\delta)}{|\mathcal{D}^{(i)}_E|}}$ and $\Delta^{(i)}_c \leq 0$.

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the difference between the policies $\pi_t^{(i)}$ and the optimal policy $\pi^*$ under the same pessimistic model $\tilde{T}_\phi^{(i)} = \arg\min_{T_\phi \in \mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t}$, makes a significant difference. Additionally, our work provides a detailed comparative analysis with [26], addressing and improving upon certain limitations in its theoretical derivations. Specifically, we rigorously clarify issues related to the approximation errors arising from the gap between the function class of $Q$ and the linear function class, thereby strengthening and extending the theoretical foundations established in [26].

Based on Lemma 3, under the umbrella of SAC in (9), we decompose $\Delta_b^{(i)}$ into three terms (i.e., $\Gamma_a^{(i)}$, $\Gamma_b^{(i)}$, and $\Gamma_c^{(i)}$) as in (12), shown at the bottom of this page, for better interpretability and to isolate the components that contribute to the performance gap in the policy.

*1) Bounding $\Gamma_a^{(i)}$:* To estimate an upper bound for $\Gamma_a^{(i)}$, we express it as the difference between two expectations. The first term, which involves the advantage function $Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - V^{(i)}_{\tilde{T}_\phi,\pi_t}(s)$, is bounded by controlling the variance of policy transitions, ensuring a predictable difference between $Q$ and $V$. The second term, based on Assumption 2 and Lemma 8, can be simplified as $w_t \cdot (\varphi_{s,a} - \mathbb{E}_{a'\sim\pi_t(\cdot|s)}[\varphi_{s,a'}])$, represents the policy gradient weighted by $\|w_t\|_2 \le W$ and the characteristic vector $\|\varphi_{s,a}\|_2 \le \varphi_{\max}$ for any $(s,a)$. By applying Hölder's inequality and Lemma 8 to the feature difference and approximating the expectation over $(s,a) \sim d^{\pi^*}_{\tilde{T}_\phi}$ as an average over all actions, we reformulate $\Gamma_a^{(i)}$ as:

$$
\begin{aligned}
\Gamma_a^{(i)} &= \mathbb{E}_{(s,a)\sim d^{\pi^*}_{\tilde{T}_\phi}} \Big[ Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - \mathbb{E}_{a'\sim\pi_t} Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a') \\
&\qquad - w_t \cdot \big(\varphi_{s,a} - \mathbb{E}_{a'\sim\pi_t(\cdot|s)}[\varphi_{s,a'}]\big) \Big] \\
&\le \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}, a\sim\pi^*} \left[ \left| Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - w_t \cdot \varphi_{s,a} \right| \right] \\
&\quad + \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}, a'\sim\pi_t} \left[ \left| w_t \cdot \varphi_{s,a'} - Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a') \right| \right] \\
&\le 2|\mathcal{A}| \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}, a\sim \mathrm{Unif}(\mathcal{A})} \left[ \left| Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - w_t \cdot \varphi_{s,a} \right| \right] \\
&= 2|\mathcal{A}| \mathbb{E}_{a\sim\mathrm{Unif}(\mathcal{A})} \Bigg[ \left( \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}} f^{(i)}(s,a) - \mathbb{E}_{s\sim d^{\pi^*}_{T}} f^{(i)}(s,a) \right) \\
&\quad + \mathbb{E}_{s\sim d^{\pi^*}_{T}} f^{(i)}(s,a) \Bigg],
\end{aligned}
$$

where $f^{(i)}(s,a) := |Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - w_t \cdot \varphi_{s,a}|$ be the absolute error term.

We analyze each component to derive an upper bound for $\Gamma_a^{(i)}$. Following the proof strategy outlined in [26] and leveraging Lemma 5, we present the detailed derivation as follows:

$$
\begin{aligned}
\Delta_G^k(H) &:= \sum_{i=1}^{I} \left[ V^{(i)}_{T,\pi^*} - \max_{0\le t\le H} V^{(i)}_{T,\pi_t} \right] \le \sum_{i=1}^{I} \left[ \frac{1}{H+1} \sum_{t=0}^{H} \left( V^{(i)}_{T,\pi^*} - V^{(i)}_{T,\pi_t} \right) \right] \\
&= \sum_{i=1}^{I} \left[ \frac{1}{H+1} \sum_{t=0}^{H} \left( V^{(i)}_{T,\pi^*} - \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi^*} + \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi^*} - \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t} + \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t} - V^{(i)}_{T,\pi_t} \right) \right] \\
&= \sum_{i=1}^{I} \Bigg[ \underbrace{\frac{1}{H+1} \sum_{t=0}^{H} \left( V^{(i)}_{T,\pi^*} - \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi^*} \right)}_{\Delta_a^{(i)}} + \underbrace{\frac{1}{H+1} \sum_{t=0}^{H} \left( \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi^*} - \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t} \right)}_{\Delta_b^{(i)}} \\
&\quad + \underbrace{\frac{1}{H+1} \sum_{t=0}^{H} \left( \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t} - V^{(i)}_{T,\pi_t} \right)}_{\Delta_c^{(i)}} \Bigg]
\end{aligned} \tag{11}
$$

$$
\begin{aligned}
\Delta_b^{(i)} &= \frac{1}{H+1} \sum_{t=0}^{H} \left( \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi^*} - \min_{T_\phi\in\mathcal{M}_\mathcal{D}} V^{(i)}_{T_\phi,\pi_t} \right) = \frac{1}{(H+1)(1-\gamma)} \sum_{t=0}^{H} \mathbb{E}_{(s,a)\sim d^{\pi^*}_{\tilde{T}_\phi}} \left( Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - V^{(i)}_{\tilde{T}_\phi,\pi_t}(s) \right) \\
&= \frac{1}{(H+1)(1-\gamma)} \sum_{t=0}^{H} \Bigg[ \underbrace{\mathbb{E}_{(s,a)\sim d^{\pi^*}_{\tilde{T}_\phi}} \left[ Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a) - V^{(i)}_{\tilde{T}_\phi,\pi_t}(s) - w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s) \right]}_{\Gamma_a^{(i)}} \\
&\quad + \underbrace{\mathbb{E}_{(s,a)\sim d^{\pi^*}_{\tilde{T}_\phi}} \left[ w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s) \right] - \mathbb{E}_{(s,a)\sim d^{\pi^*}_{T_\phi}} \left[ w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s) \right]}_{\Gamma_b^{(i)}} + \underbrace{\mathbb{E}_{(s,a)\sim d^{\pi^*}_{T_\phi}} \left[ w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s) \right]}_{\Gamma_c^{(i)}} \Bigg]
\end{aligned} \tag{12}
$$

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- We note that as $a \sim \text{Unif}(\mathcal{A})$, $\mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}, a\sim\text{Unif}(\mathcal{A})} f^{(i)}(s,a) = \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}}[\frac{1}{|\mathcal{A}|}\sum_a f^{(i)}(s,a)]$. By defining an induced MDP with a modified reward function $\tilde{r}^{(i)}(s,a) := \tilde{f}^{(i)}(s) = \frac{1}{|\mathcal{A}|}\sum_a f^{(i)}(s,a)$, which depends only on $s$ and remains the same for all $a$, we obtain:

$$\begin{aligned}\mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}} \tilde{f}^{(i)}(s) &= \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}} \tilde{r}^{(i)}(s,a) \\ &= \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}, a\sim\pi^*} \tilde{r}^{(i)}(s,a).\end{aligned}$$

We derive an upper bound on the expected value difference by state-action trajectory sampling. Specifically, we have the bound $\tilde{f} \le |W \cdot \|\varphi_{s,a}\|_{2,d^{\pi}_{T\circ\text{Unif}}} + R_{\max}/(1-\gamma)|$, where $\|\varphi_{s,a}\|_{2,d^{\pi}_{T\circ\text{Unif}}}$ denotes the $L_2$ -norm of the feature under the distribution $d^{\pi}_{T\circ\text{Unif}}$. Thus, we can establish the following upper bound for the value function induced by the modified reward function $\tilde{r}^{(i)}$ as:

$$\tilde{V} \le \frac{1}{1-\gamma}\cdot\left|W\cdot\|\varphi_{s,a}\|_{2,d^{\pi}_{T\circ\text{Unif}}} + \frac{R_{\max}}{1-\gamma}\right|.$$

Hence, based on Definition 2 and Lemma 5, we obtain:

$$\begin{aligned}&\mathbb{E}_{\substack{s\sim d^{\pi^*}_{\tilde{T}_\phi}\\ a\sim\text{Unif}(\mathcal{A})}} f^{(i)}(s,a) - \mathbb{E}_{\substack{s\sim d^{\pi^*}_{T}\\ a\sim\text{Unif}(\mathcal{A})}} f^{(i)}(s,a) \\ &= \mathbb{E}_{s\sim d^{\pi^*}_{\tilde{T}_\phi}}\frac{1}{|\mathcal{A}|}\sum_a f^{(i)}(s,a) - \mathbb{E}_{s\sim d^{\pi^*}_{T}}\frac{1}{|\mathcal{A}|}\sum_a f^{(i)}(s,a) \\ &\le \frac{2\gamma}{(1-\gamma)^2}\left[W\cdot\|\varphi_{s,a}\|_{2,d^{\pi}_{T\circ\text{Unif}}} + \frac{R_{\max}}{1-\gamma}\right] \\ &\quad\cdot\sqrt{C_{\pi^*}}\mathbb{E}_{(s,a)\sim\rho}\left[\text{TV}\left(\tilde{T}_\phi(\cdot\mid s,a), T(\cdot\mid s,a)\right)^2\right] \\ &\le \frac{2\gamma}{(1-\gamma)^2}\left[W\cdot\|\varphi_{s,a}\|_{2,d^{\pi}_{T\circ\text{Unif}}} + \frac{R_{\max}}{1-\gamma}\right] \\ &\quad\cdot\sqrt{\frac{C_{\pi^*}\ln\left(c_2|\mathcal{M}_\mathcal{D}|/\delta\right)}{|\mathcal{D}^{(i)}_E|}} \\ &\le \left[\frac{2\gamma c_3}{(1-\gamma)^2} + \frac{2\gamma R_{\max}}{(1-\gamma)^3}\right]\cdot\sqrt{\frac{C_{\pi^*}\ln\left(c_2|\mathcal{M}_\mathcal{D}|/\delta\right)}{|\mathcal{D}^{(i)}_E|}}.\end{aligned} \tag{13}$$

where $W$ is the upper bound on the weight norm, $|\mathcal{D}^{(i)}_E|$ is the sample size of the dataset $\mathcal{D}^{(i)}_E$, and $c_3 = W\cdot\varphi_{\max}$.

- To analyze the second part of $\Gamma^{(i)}_a$, we establish a relationship between the distributions induced by the two policies by expressing it through their density ratio. Applying Lemma 7, we bound the expectation over the uniform action distribution and the stationary distribution of $\pi^*$. Using the triangle inequality, we obtain

$$\begin{aligned}&\mathbb{E}_{s\sim d^{\pi^*}_T, a\sim\text{Unif}(\mathcal{A})}[f^{(i)}(s,a)] \\ &\le \left\|\frac{d^{\pi^*}_T(s)}{d^{\pi_t}_{\tilde{T}_\phi}(s)}\right\|^{\frac{1}{2}}_\infty \sqrt{\mathbb{E}_{s\sim d^{\pi_t}_{\tilde{T}_\phi}, a\sim\text{Unif}(\mathcal{A})}\left[f^{(i)}(s,a)^2\right]} \\ &\le \left\|\frac{d^{\pi^*}_T}{d^{\pi_t}_{\tilde{T}_\phi}}\right\|^{\frac{1}{2}}_\infty \sqrt{2}\sqrt{\mathbb{E}_{s,a}\left[\left|Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a)\right|^2\right] + c_3^2} \\ &\le \left\|\frac{2d^{\pi^*}_T}{d^{\pi_t}_{\tilde{T}_\phi}}\right\|^{\frac{1}{2}}_\infty \left[\sqrt{\mathbb{E}_{s,a}\left[\left|Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a)\right|^2\right]} + c_3\right],\end{aligned} \tag{14}$$

where the last inequality comes directly from applying the generalized triangle inequality. Subsequently, we apply Hoeffding's inequality together with Lemma 6 to quantify the generalization error as follows:

$$\begin{aligned}&\sqrt{\mathbb{E}_{s,a}\left[\left|Q^{(i)}_{\tilde{T}_\phi,\pi_t}(s,a)\right|^2\right]} \\ &\le \sqrt{\sum_{(s,a)\in\mathcal{D}^{(i)}_E}\frac{|Q^{(i)}(s,a)|^2}{\left|\mathcal{D}^{(i)}_E\right|}} + \frac{R_{\max}}{1-\gamma}\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}^{(i)}_E\right|}\right)^{1/4} \\ &\le \left[\frac{\left|\mathcal{D}^{(i)}_E\right|\cdot\left(\frac{R_{\max}}{1-\gamma}\right)^2}{\left|\mathcal{D}^{(i)}_E\right|}\right]^{1/2} + \frac{R_{\max}}{1-\gamma}\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}^{(i)}_E\right|}\right)^{1/4} \\ &= \frac{R_{\max}}{1-\gamma} + \frac{R_{\max}}{1-\gamma}\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}^{(i)}_E\right|}\right)^{1/4}.\end{aligned} \tag{15}$$

By merging (14) with (15), we obtain an upper bound for the second term of $\Gamma^{(i)}_a$ as shown in (16). Unlike [26], our approach leverages empirical estimation and a rigorous generalization error analysis based on Hoeffding's inequality to characterize the gap between the function class of $Q(s,a)$ and the linear function class $w_t\cdot\varphi_{s,a}$. With high probability, we have:

$$\begin{aligned}&\mathbb{E}_{s\sim d^{\pi^*}_T, a\sim\text{Unif}(\mathcal{A})}[f^{(i)}(s,a)] \\ &\lesssim \frac{\sqrt{2}}{1-\gamma}\left\|\frac{d^{\pi^*}_T}{\mu_0}\right\|^{\frac{1}{2}}_\infty\left[\frac{R_{\max}}{1-\gamma}\left(1+\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}^{(i)}_E\right|}\right)^{1/4}\right) + c_3\right].\end{aligned} \tag{16}$$

Thus, by combining these results, we can upper-bound the original term by summing up the contributions from the two parts, (13) and (16), we have:

$$\begin{aligned}\Gamma^{(i)}_a \le\ &\left[\frac{4\gamma c_3|\mathcal{A}|}{(1-\gamma)^2} + \frac{4\gamma|\mathcal{A}|R_{\max}}{(1-\gamma)^3}\right]\sqrt{\frac{C_{\pi^*}\ln\left(c_2|\mathcal{M}_\mathcal{D}|/\delta\right)}{|\mathcal{D}^{(i)}_E|}} \\ &+ \frac{2\sqrt{2}|\mathcal{A}|}{1-\gamma}\left\|\frac{d^{\pi^*}_T}{\mu_0}\right\|^{\frac{1}{2}}_\infty \\ &\times\left[\frac{R_{\max}}{1-\gamma} + \frac{R_{\max}}{1-\gamma}\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}^{(i)}_E\right|}\right)^{1/4} + c_3\right].\end{aligned} \tag{17}$$

*2) Bounding $\Gamma^{(i)}_b$:* Notably, $\Gamma^{(i)}_b$ measures the difference between two expectations over the visitation measures under

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transition models $\tilde{T}_\phi^{(i)}$ and $T$, that is,

$$\Gamma_b^{(i)} = \mathbb{E}_{(s,a)\sim d_{\tilde{T}_\phi}^{\pi^*}}\left[w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s)\right] - \mathbb{E}_{(s,a)\sim d_T^{\pi^*}}\left[w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s)\right].$$

Using trajectories generated by $\pi^*$ under the distributions $\tilde{T}_\phi^{(i)}$ and $T$, respectively, we can introduce the another reward function $\tilde{r}^{(i)}(s,a) := w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s)$. Based on Assumption 2 and Lemma 8, we derive the following upper bound for $\tilde{r}^{(i)}(s,a)$ as:

$$\begin{aligned}\left|\tilde{r}^{(i)}(s,a)\right| &= \left|w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a \mid s)\right| \\ &= \|w_t\|_2 \cdot \left\|\varphi_{s,a} - \mathbb{E}_{a'\sim\pi_t^{(i)}(\cdot|s)}\varphi_{s,a'}\right\| \\ &\le 2W \cdot \varphi_{\max} = 2c_3.\end{aligned}$$

Consequently, the value function is bounded by $|\tilde{V}^{(i)}_{T_\phi,\pi_t}(s)| \le \frac{2c_3}{1-\gamma}$ for all $s \in \mathcal{S}$.

Similar to the bound of $\Gamma_a^{(i)}$, we can incorporate Lemma 1 and Lemma 5 to bound $\Gamma_b^{(i)}$ in terms of the total variation distance between $\tilde{T}_\phi^{(i)}$ and $T$, and finally obtain:

$$\begin{aligned}\Gamma_b^{(i)} &= \mathbb{E}_{\tau\sim\tilde{T}_\phi^{(i)}}\left[\sum_{t=0}^{H}\gamma^t\tilde{r}^{(i)}(s_t,a_t)\right] - \mathbb{E}_{\tau\sim T}\left[\sum_{t=0}^{H}\gamma^t\tilde{r}^{(i)}(s_t,a_t)\right] \\ &= \left|\tilde{V}^{(i)}_{\tilde{T}_\phi,\pi_t} - \tilde{V}^{(i)}_{T,\pi_t}\right| \\ &\le \frac{4c_3\gamma}{(1-\gamma)^2}\mathbb{E}_{(s,a)\sim d^{\pi_t}_{\tilde{T}_\phi}}\left[\mathrm{TV}(\tilde{T}_\phi^{(i)}(\cdot|s,a), T(\cdot|s,a))\right] \\ &\le 4c_3\gamma(1-\gamma)^{-2}\sqrt{\frac{C_{\pi^*}\ln\left(c_2\left|\mathcal{M}_\mathcal{D}\right|/\delta\right)}{\left|\mathcal{D}_E^{(i)}\right|}}. \end{aligned} \tag{18}$$

*3) Bounding $\Gamma_c^{(i)}$:* We start from the definition of $\Gamma_c^{(i)} = \mathbb{E}_{(s,a)\sim d_T^{\pi^*}}[w_t \cdot \nabla_\psi \log \pi_t^{(i)}(a|s)]$. By Lemma 10, $\log \pi_t^{(i)}(a|s)$ is $\varphi_{\max}^2$-smooth. By substituting the gradient update $\psi_{t+1} - \psi_t = \eta w_t$ and applying the smoothness property in Lemma 10, we can have:

$$\begin{aligned}\Gamma_c^{(i)} &= \frac{1}{\eta}\mathbb{E}_{(s,a)\sim d_T^{\pi^*}}\left[\nabla_\psi \log \pi_t^{(i)}(a|s)\cdot(\psi_{t+1}-\psi_t)\right] \\ &\le \frac{1}{\eta}\mathbb{E}_{(s,a)\sim d_T^{\pi^*}}\left[\log \pi_{t+1}^{(i)}(a|s) - \log \pi_t^{(i)}(a|s)\right] \\ &\quad + \frac{\varphi_{\max}^2}{2\eta}\|\psi_{t+1}-\psi_t\|^2 \\ &\le \frac{1}{\eta}\mathbb{E}_{(s,a)\sim d_T^{\pi^*}}\left[\log \pi_{t+1}^{(i)}(a|s) - \log \pi_t^{(i)}(a|s)\right] + \frac{c_3^2\eta}{2} \\ &= \frac{1}{\eta}\mathbb{E}_{(s,a)\sim d_T^{\pi^*}}\left[\mathrm{D_{KL}}\left(\pi_{t+1}^{(i)}(\cdot|s)\|\pi^*(\cdot|s)\right)\right. \\ &\quad \left. - \mathrm{D_{KL}}\left(\pi_t^{(i)}(\cdot|s)\|\pi^*(\cdot|s)\right)\right] + \frac{c_3^2\eta}{2}.\end{aligned}$$

Taking the expectation of $\Gamma_c^{(i)}$ and averaging over $t = 0,\ldots,H$, we arrive at the following bound:

$$\begin{aligned}\frac{1}{H+1}\sum_{t=0}^{H}\Gamma_c^{(i)} &\le \frac{\mathbb{E}_{s\sim d_T^{\pi^*}}\left[\mathrm{D_{KL}}\left(\pi_0^{(i)}(\cdot|s)\|\pi^*(\cdot|s)\right)\right]}{\eta(H+1)} + \frac{c_3^2\eta}{2} \\ &\le \frac{c_3^2\eta}{2} + \frac{\log|\mathcal{A}|}{\eta(H+1)},\end{aligned} \tag{19}$$

where we take account of the initial policy $\pi_0(\cdot|s) = \mathrm{Uniform}(\mathcal{A})$, and correspondingly have $\mathrm{D_{KL}}(\pi_0\|\pi^*) \le \log|\mathcal{A}|$.

By combining (17), (18), and (19), we can derive the following upper bound for $\Delta_b^{(i)}$:

$$\begin{aligned}\Delta_b^{(i)} &\le \frac{1}{(H+1)(1-\gamma)}\sum_{t=0}^{H}\left[\Gamma_a^{(i)} + \Gamma_b^{(i)} + \Gamma_c^{(i)}\right] \\ &\le \left[\frac{4c_3\gamma\left(|\mathcal{A}|+1\right)}{(1-\gamma)^3} + \frac{4\gamma|\mathcal{A}|R_{\max}}{(1-\gamma)^4}\right]\sqrt{\frac{C_{\pi^*}\ln\left(c_2\left|\mathcal{M}_\mathcal{D}\right|/\delta\right)}{\left|\mathcal{D}_E^{(i)}\right|}} \\ &\quad + \frac{2\sqrt{2}|\mathcal{A}|}{(1-\gamma)^2}\left\|\frac{d_T^{\pi^*}}{\mu_0}\right\|_\infty^{\frac{1}{2}}\cdot\left[\frac{R_{\max}}{1-\gamma} + \frac{R_{\max}}{1-\gamma}\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}_E^{(i)}\right|}\right)^{1/4} + c_3\right] \\ &\quad + \frac{c_3^2\eta}{2(1-\gamma)} + \frac{\log|\mathcal{A}|}{\eta(H+1)(1-\gamma)}.\end{aligned}$$

Thus, the summarized upper bounds $\Delta_a^{(i)}$, $\Delta_b^{(i)}$ and $\Delta_c^{(i)}$ for the PAC bound of a single agent are given in the following Lemma.

*Lemma 2:* Let $\eta$ and $\xi$ be some constant such that $T \in \mathcal{M}_\mathcal{D}$ holds with high probability at least $1-\delta$. Suppose that the characteristic vector and policy gradient weights are bounded, i.e., $\|\varphi_{s,a}\|_2 \le \varphi_{\max}$ and $\|w_t\|_2 \le W$, and define $c_3 = W \cdot \varphi_{\max}$. Then under Assumptions 1 and 2, there exist positive constants $c_1$ and $c_2$ such that:

$$\begin{aligned}V_{T,\pi^*}^{(i)} - \max_{0\le t\le H} V_{T,\pi_t}^{(i)} &\le \sqrt{\frac{C_{\pi^*}\ln\left(c_2\left|\mathcal{M}_\mathcal{D}\right|/\delta\right)}{\left|\mathcal{D}_E^{(i)}\right|}} \\ &\quad \cdot\left[\frac{c_1}{(1-\gamma)^2} + \frac{4c_3\gamma\left(|\mathcal{A}|+1\right)}{(1-\gamma)^3} + \frac{4\gamma|\mathcal{A}|R_{\max}}{(1-\gamma)^4}\right] \\ &\quad + \frac{2\sqrt{2}|\mathcal{A}|}{(1-\gamma)^2}\left\|\frac{d_T^{\pi^*}}{\mu_0}\right\|_\infty^{\frac{1}{2}}\cdot\left[\frac{R_{\max}}{1-\gamma} + \frac{R_{\max}}{1-\gamma}\left(\frac{\ln(2/\delta)}{2\left|\mathcal{D}_E^{(i)}\right|}\right)^{1/4} + c_3\right] \\ &\quad + \frac{c_3^2\eta}{2(1-\gamma)} + \frac{\log|\mathcal{A}|}{\eta(H+1)(1-\gamma)}.\end{aligned}$$

*Remark:* In our theory, $C_{\pi^*}$ serves only as a *transfer* constant: if errors such as TV discrepancies are controlled under $\rho$, they are also controlled under $d_T^{\pi^*}$. Meanwhile, Pinsker's inequality [47] bridges the relationship between measurable and operational KL divergence and the TV distance, providing a computationally efficient upper bound to quantify data–policy mismatch and to drive conservative or safety adaptations.

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### C. Group PAC Bounds

We assume all CAVs with constrained communication range can form a graph structure. Specifically, for each episode $k$, a neighborhood graph, denoted as $\mathcal{G}_{d,k}^{(i)} = (\mathcal{F}_{d,k}^{(i)}, \mathcal{E}_{d,k}^{(i)}) \subset \mathcal{G}$, is constructed. Here, $\mathcal{E}_{d,k}^{(i)}$ is the set of communication links between agent $i$ and other agents $i' \in \mathcal{F}_{d,k}^{(i)}$. We define the total number of samples $N_{d,k}^{(i)}$ for vehicle $i$ after communication as the cumulative sum of observations collected from itself and its neighbors within the communication range $d \in [0, D(\mathcal{G}_k)]$. Therefore, at the end of episode $k$,

$$N_{d,k}^{(i)} := N_{d,k-1}^{(i)} + \sum\nolimits_{i' \in \mathcal{G}_{d,k}^{(i)}} |\mathcal{D}_{E,k}^{(i')}|,$$

where $\mathcal{D}_{E,k}^{(i')}$ represents the data collected by vehicle $i$ during its interactions with the environment in episode $k$.

To address the challenge of directly estimating $N_{d,k}^{(i)}$, we introduce the concept of a clique cover [49] for $\mathcal{F}_{d,k}^{(i)}$, which is a collection of cliques that collectively cover all vertices of a power graph. Let $\mathbf{C}_{d,k}$ denote a clique cover of $\mathcal{G}_{d,k}$, where the graph contains multiple maximal cliques $\mathcal{C}_{d,k}^1, \mathcal{C}_{d,k}^2, \ldots, \mathcal{C}_{d,k}^{I'} \in \mathbf{C}_{d,k}$ with $I' \leq I$. The size of each clique $\mathcal{C}_{d,k}^{i'}$ is denoted by $|\mathcal{C}_{d,k}^{i'}|$. For each clique $\mathcal{C}_{d,k}^{i'}$, let $N_{\mathcal{C},k}^{i'}$ represent the total number of samples exchanged within the clique $\mathcal{C}_{d,k}^{i'}$ at the end of the episode $k$, which are accessible to all vehicles $i \in \mathcal{C}_{d,k}^{i'}$. Intuitively, the clique term reflects the communication-range–limited information exchange among vehicles, where each clique represents a group of vehicles that can mutually communicate and share collected samples. To simplify the computation, we approximate $N_{\mathcal{C},k}^{i'} \approx |\mathcal{C}_{d,k}^{i'}| kH$, which satisfies the definition. The clique cover number $\bar{\chi}(\mathcal{G}_{d,k})$ is defined as the minimum number of cliques required to cover the power graph $\mathcal{G}_{d,k}$ under the communication range. Furthermore, we define $\bar{\chi}(\mathcal{G}_d) := \max_{k \in [K]} \{\bar{\chi}(\mathcal{G}_{d,k})\}$ as the maximum clique cover number across all episodes.

Thus, by utilizing the definition of cliques and bounding the contributions within each clique according to Lemma 11, we can have

$$\sum_{i=1}^{I} \frac{1}{\sqrt{|\mathcal{D}_E^{(i)}|}} = \sum_{i'=1}^{I'} \sum_{i \in \mathcal{C}_{d,k}^{i'}} \frac{1}{\sqrt{N_{d,k}^{(i)}}} \leq \sum_{i'=1}^{I'} |\mathcal{C}_{d,k}^{i'}| \frac{1}{\sqrt{N_{\mathcal{C},k}^{i'}}} \lesssim \sqrt{\bar{\chi}(\mathcal{G}_d)\, I / kH}.$$

We now present Theorem 1, which provides a theoretical guarantee for the multi-agent system, establishing an upper bound on the sub-optimality of the optimal policy among the iterated policies in MA-PMBRL 1 at the end of each episode.

*Theorem 1 (PAC bound for* MA-PMBRL*):* Let $\eta$ and $\xi$ be constants chosen such that $T \in \mathcal{M}_{\mathcal{D}}$ holds with probability at least $1 - \delta$. Assume the policy gradient satisfies $\|w_t\|_2 \leq W$, and the feature vector satisfies $\|\varphi_{s,a}\|_2 \leq \varphi_{\max}$. Define constants $c_1, c_2$ and $c_3 = W \cdot \varphi_{\max}$. Then, at the end of the $k$-th episode with a maximum length $H$, the algorithm's performance gap:

$$\Delta_G^k(H) = \sum_{i=1}^{I} \left[ \frac{1}{H+1} \sum_{t=0}^{H} \left( V_{T,\pi^*}^{(i)} - V_{T,\pi_t}^{(i)} \right) \right],$$

satisfies the following bound:

$$\begin{aligned} \Delta_G^k(H) \leq & \left[ \frac{c_1}{(1-\gamma)^2} + \frac{4c_3\gamma\,(|\mathcal{A}|+1)}{(1-\gamma)^3} + \frac{4\gamma|\mathcal{A}|R_{\max}}{(1-\gamma)^4} \right] \\ & \cdot \sqrt{C_{\pi^*}\bar{\chi}\,(\mathcal{G}_d)\, I \ln\left(c_2\,|\mathcal{M}_{\mathcal{D}}|\,/\delta\right)/kH} \\ & + \frac{2\sqrt{2}|\mathcal{A}|}{(1-\gamma)^2} \left\| \frac{d_T^{\pi^*}}{\mu_0} \right\|_\infty^{\frac{1}{2}} \left[ \frac{R_{\max}}{1-\gamma} \sqrt[4]{\frac{\ln(\frac{2}{\delta})\bar{\chi}\,(\mathcal{G}_d)\, I}{2kH}} \right. \\ & \left. + \frac{IR_{\max}}{1-\gamma} + c_3 I \right] + \frac{c_3^2 \eta I}{2(1-\gamma)} + \frac{I \log|\mathcal{A}|}{\eta(H+1)(1-\gamma)}. \end{aligned} \tag{20}$$

*Remark:* In summary, Theorem 1 establishes a *finite-sample, group-level* PAC guarantees bound in (20) for multi-agent learning under partial coverage $C_{\pi^*} < \infty$ and constrained communication, holding with probability at least $1 - \delta$ under a realizable hypothesis class.

The analysis mirrors the algorithmic modules in Section IV: MLE dynamics learning ((5)), construction of a TV-based confidence set ((6)), and *worst-case* pessimistic model selection via PGD ((7) and (8)) in Section IV. It also includes error decomposition ((11)) under the selected model and decentralized policy updates with short-horizon rollouts (Algorithm 1, (9)). Theorem 1 further bounds the cumulative sub-optimality by showing that, even with pessimistic model selection (8), the performance gap $\Delta_G^k(H)$ remains controlled. The bound scales with data scarcity and the horizon $H$, reflecting larger model and planning errors under limited coverage and longer horizons. Unlike non-communicative settings where group guarantees typically add single-agent bounds, our result depends on the network structure through the clique-cover number $\chi(\mathcal{G}_{\Delta_k})$ under range-$d$ communication, indicating that structured neighborhood communication can reduce sample complexity relative to isolated agents and accelerate MARL convergence.

TABLE II
ORDER-LEVEL COMPUTATIONAL COMPLEXITY COMPARISON

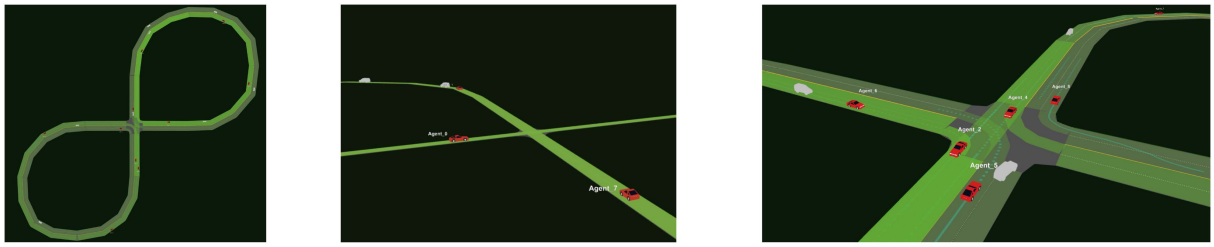
| Algorithm | Training-time complexity | Execution-time complexity |
|---|---|---|
| MA-PMBRL | $\mathcal{O}(KIH(L_{\text{rollout}} + X))$ | $\mathcal{O}(I)$ |
| MA-PETS | – | $\mathcal{O}(IHJQH)$ |
| MBPO | $\mathcal{O}(KIH\,L_{\text{rollout}})$ | $\mathcal{O}(I)$ |

### D. Computation Complexity Analysis

Here, we briefly analyze the order-level computational complexity of MA-PMBRL, MA-PETS, and MBPO using the notations summarized in Table I.

Table II reports the resulting training-time and execution-time complexity for MA-PMBRL and representative baselines. Notably, the PGD procedure introduces an additional overhead of $X$ projected first-order updates for each agent. Since

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(a) "Figure Eight" loop (b) Single-lane "Unprotected Intersection" (c) Multi-lane "Unprotected Intersection"

Fig. 3. The "Unprotected Intersection" scenario in the closed single-lane and multi-lane "Figure Eight" loop for Simulations. (a) presents an aerial view of the "Figure Eight" loop, while (b) and (c) provide the regional enlarged view of the single-lane and multi-lane "Unprotected Intersection", respectively.

MA-PMBRL is decentralized and communication-range constrained, this overhead mainly depends on local neighboring interactions within range $d$, rather than the full global joint state-action space, thereby supporting scalable implementation in large-scale multi-agent systems. In practice, the maximum number of PGD iterations for pessimistic model selection $X$ is chosen as a small constant (i.e., $X = 6$ in our simulations), and the resulting additional overhead remains bounded. Detailed derivations are deferred to Appendix C. In addition to achieving the best task performance in our experiments, MA-PMBRL incurs only a moderate extra training-time overhead of $\mathcal{O}(KIHX)$ over MBPO due to the PGD-based pessimistic model selection, while retaining the same order of execution-time complexity. By comparison, MA-PETS exhibits the largest execution-time complexity because it relies on online planning at each decision step.

## VI. Experimental Settings and Numerical Results

### A. Experimental Settings

In this section, we present comprehensive performance evaluations of our proposed MA-PMBRL framework within autonomous vehicle control applications, systematically comparing its effectiveness against leading RL approaches, including MA-PETS [32], MBPO [18], FIRL [8], SVMIX [9], and SAC [27]. The experimental validation is conducted using the SMARTS [50] simulation platform, specifically employing the challenging "Unprotected Signal-Free Intersection" scenario within a closed-loop single-lane and multi-lane "Figure Eight" traffic configuration, which is a representative mixed-autonomy environment depicted in Fig. 3.

Our experimental setup explicitly includes mixed traffic, consisting of $I = 8, 22, 50$ CAVs controlled by MA-PMBRL, together with a random number of human-driven vehicles (HVs), denoted by $I_{\text{hv}} \in [8, 22]$, as surrounding traffic participants. All CAVs initiate navigation from a unidirectional approach lane, continuously traversing the intersection while optimizing for collision avoidance, congestion mitigation, and traffic flow efficiency. The vehicular interactions strictly adhere to the MDP formulation detailed in Section III, with specific state-action-reward mappings tailored for the "Unprotected Intersection" scenario as follows:

- *Observable State Space:* Each CAV maintains a partial observation field limited to its kinematic status and proximate vehicular context. As described in Section III-B, the perceptual state for agent $i$ at time $t$ comprises:

$$s_t^{(i)} = \left( \underbrace{v_t^{(i)}, x_t^{(i)}, y_t^{(i)}}_{\substack{\text{ego} \\ \text{kinematics}}}, \underbrace{v_{t,a}^{(i)}, l_{t,a}^{(i)}}_{\substack{\text{leading} \\ \text{vehicle}}}, \underbrace{v_{t,e}^{(i)}, l_{t,e}^{(i)}}_{\substack{\text{following} \\ \text{vehicle}}}, b_t^{(i)} \right) \in \mathbb{R}^8 \times \{0,1\}.$$

  Notably, $b_t^{(i)}$ functions as a collision risk flag triggered when any vehicle enters the emergency braking radius ($\|l_{t,a/e}\| < l_{\text{safe}}$).
- *Control Actions:* The action space for each CAV integrates longitudinal control with lane change decisions:

$$a_t^{(i)} = \left( \underbrace{v_t^{(i)}}_{\substack{\text{target speed} \\ \text{(continuous)}}}, \underbrace{c_t^{(i)}}_{\substack{\text{lane change} \\ \text{(binary)}}} \right) \in \mathbb{R} \times \{0,1\}$$

  In particular, in single-lane configurations, the lane-change component $c_t^{(i)}$ is inactive (i.e., $c_t^{(i)} \equiv 0$), so the action reduces only to the speed regulation.
- *Cooperative Reward Design:* The reward function is designed to encourage collision-free speed maximization while implicitly coordinating traffic flow through observations of neighboring vehicle speeds. Specifically, it is formulated as:

$$r_t^{(i)} = \underbrace{v_t^{(i)}}_{\substack{\text{ego speed} \\ \text{optimization}}} + \lambda \left( \underbrace{v_{t,a}^{(i)} + v_{t,e}^{(i)}}_{\substack{\text{neighbor vehicle} \\ \text{speed coordination}}} \right) - \zeta \cdot \mathbb{I}(b_t^{(i)})$$

  where $\lambda$ regulates the influence of neighboring vehicle speeds to facilitate coordinated traffic flow, and $\zeta$ penalizes emergency braking events to enhance both individual efficiency and emergent collective behavior. By incorporating observable neighbor speeds, the reward function fosters a balance between local optimization and global coordination.

Given that directly summing rewards across multiple agents as defined above may introduce computational redundancy and

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TABLE III
PARAMETER CONFIGURATIONS FOR THE SMARTS SIMULATOR AND MA-PMBRL FRAMEWORK

| Parameter | Value |
|---|---|
| **SMARTS Simulator** | |
| Discrete-time step | $\tau = 0.1$s |
| No. of vehicles | $I \in \{8, 22, 50\}$ |
| No. of human-driven vehicles | $I_{\text{hv}} \in [8, 22]$ |
| "Figure Eight" track length | 480m |
| Speed limits (min, max) | $0 \leq v \leq 13.89$ m/s |
| **MA-PMBRL Framework** | |
| Speed observation | $v_t \in [0, 13.89]$ m/s |
| Position coordinates | $z_t \in \mathbb{R}^2$ |
| Target speed | $\bar{v}_t \in [0, 13.89]$ m/s |
| Speed of nearby vehicles | $v_{t,a}, v_{t,e} \in [0, 13.89]$ m/s |
| Distance to nearby vehicles | $l_{t,a}, l_{t,e} \in [0, 75]$ m |
| Emergency braking indicator | $b^{(i)} \in \{0, 1\}$ |
| Travel distance per vehicle | $\Lambda x^{(i)} \in \mathbb{R}$ |
| No. of episodes | $K = 20$ |
| Max episode length | $H = 1500$ |
| Communication range | $d \in [0, 200]$ m |
| Model rollout length | $L_{\text{rollout}} = 8$ |
| Reward weight | $\lambda = 0.85$ |
| Safety penalty | $\zeta = 7.5$ |
| **Learning Rates** | |
| PGD | $\alpha_{\text{PGD}} = 0.001$ |
| Actor network | $\alpha_{\text{actor}} = 0.0001$ |
| Critic network | $\alpha_{\text{critic}} = 0.001$ |
| Temperature parameter | $\alpha_{\text{temp}} = 0.001$ |
| **Training Parameters** | |
| Real-to-model data ratio | 0.7 |
| Discount factor | $\gamma = 0.98$ |
| Soft update parameter | 0.01 |
| Random seed | 5 |
| **Probabilistic Deep Neural Network** | |
| Replay buffer size | $20,000$ |
| Model buffer size | $7,000$ |
| No. of hidden layers | 3 |
| No. of hidden units | 256 |
| Training batch size | $1,000$ |
| Mini-batch size | 512 |
| Optimizer | Adam |

* Parameters specific to MA-PMBRL.
† Temperature parameter ($\alpha_{\text{temp}}$) controls entropy regularization in maximum entropy RL.

evaluation bias due to overlapping vehicle states, we propose a novel system-level utility metric grounded in traffic efficiency analysis. By incorporating the cumulative travel distance $\Lambda x_t^{(i)}$ of vehicle $i$ until the last time-step of episode $k$, the system utility metric is formulated as:

$$\text{Utility}(K) = \mathbb{E}_{\pi \sim \tilde{T}_{\phi_k}} \frac{1}{HI} \sum\nolimits_{i=1}^{I} \Lambda x_H^{(i)} \tag{21}$$

where $\tilde{T}_{\phi_k}$ ($\forall k \in [K]$) denotes the updated dynamics model until the episode $k$ with 20 total training episodes and $H = 1500$ maximum collision-free time-steps per episode in single-lane scenarios. This metric mitigates state redundancy in naive reward aggregation while quantifying global traffic efficiency through normalized cumulative travel distances, thereby ensuring robust evaluation of multi-vehicle coordination. Finally, all experiments are conducted on the NVIDIA GeForce RTX 4090 with 5 independent simulations, with full parameter specifications for the traffic environment, DNN architectures, and MA-PMBRL hyperparameters detailed in Table III.

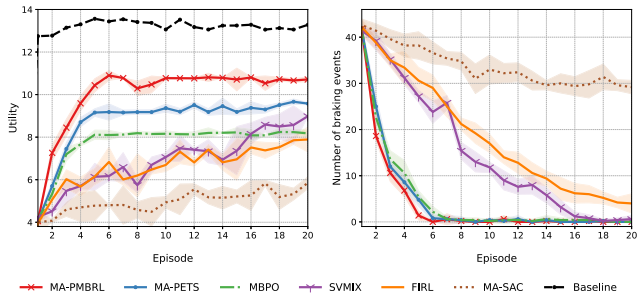



Fig. 4. Comparison of utility in the single-lane "Unprotected Intersection" scenario.

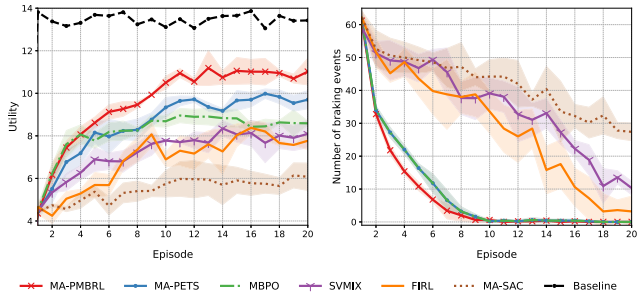



Fig. 5. Comparison of utility in the multi-lane "Unprotected Intersection" scenario.

## B. Numerical Results

*1) Validation of Performance Superiority:* We evaluate the performance of MA-PMBRL for CAVs with $I = 8$ and communication range $d = 100$, comparing it with other state-of-the-art MBRL and MFRL algorithms such as MA-PETS, MBPO, SVMIX, MA-SAC, and FIRL in Fig. 4. Furthermore, an optimal baseline is established using SMARTS, where all vehicles are controlled based on a typical car-following model that incorporates significant prior knowledge. From the *utility* curves (left panel), MA-PMBRL achieves superior utility over the episodes, in terms of faster convergence and more consistent improvements throughout the training process. Additionally, the *number of braking events* (right panel) shows that MA-PMBRL rapidly drives the count to near zero within a few episodes and keeps it low thereafter, whereas SVMIX and FIRL decay more gradually and MA-SAC remains persistently elevated. This reduction of unnecessary braking indicates smoother control and improved safety under the same traffic conditions. Overall, these results make MA-PMBRL a strong choice for scenarios requiring both high coordination efficiency and robust, stable driving behavior.

Building on the single-lane results, we consider a more challenging setting: a multi-lane unprotected intersection with $I = 22$ CAVs and communication range $d = 100$ (Fig. 5). Fig. 5 (left) demonstrates that MA-PMBRL attains high utility with fast and stable convergence. However, its collision-avoidance curve is slightly more conservative in the very early episodes than MAPPO, SVMIX, SAC. Crucially, Fig. 5 (right) shows that MA-PMBRL drives the number of braking events down rapidly, reaching near-zero within the first 6–8 episodes and remaining low thereafter, while other MFRL decay more

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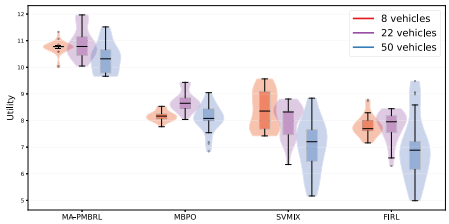



Fig. 6. Comparison of utility across different traffic densities under the Unprotected "Figure Eight" loop scenario.

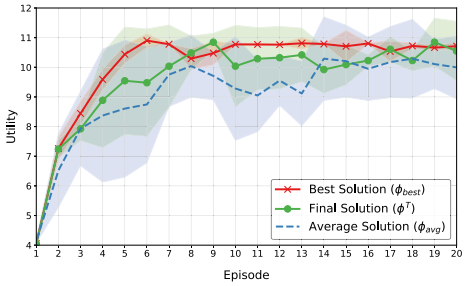



Fig. 7. Comparison of the utility of different choices of PGD in the single-lane "Unprotected Intersection" scenario.

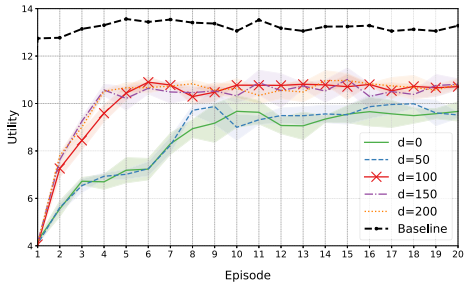



Fig. 8. Performance comparison under different communication range $d$.

TABLE IV
THE COMMUNICATION OVERHEADS OF DIFFERENT COMMUNICATION RANGE $d$

| Name | Value | | | | |
|---|---|---|---|---|---|
| Communication Distance $d$ | 0 | 50 | 100 | 150 | 200 |
| Communication Overheads | 0 | 241.9 | 1,443.7 | 5,635.3 | 7,192.5 |

gradually or stay elevated. The substantial reduction of unnecessary braking indicates smoother, less stop-and-go trajectories and complements the utility gains. Overall, MA-PMBRL provides fast utility convergence, strong safety performance, and markedly reduced braking in dense, signal-free multi-lane intersections. To further assess robustness under mixed traffic, we additionally vary the number of surrounding HVs in the multi-lane setting with $I = 22$ CAVs. The results show that MA-PMBRL remains stable under increasing HV density, with only moderate utility degradation as the surrounding traffic becomes more congested; detailed settings and results are provided in Appendix D.1.

To evaluate scalability to denser traffic and better reflect practical deployments with higher traffic density, crowded intersections, we deploy the policy learned in the multi-lane setting with $I = 22$ CAVs to a more congested multi-lane scenario with $I = 50$ CAVs in a zero-shot manner without additional fine-tuning, as shown in Fig. 6. For reference, we also include the in-distribution evaluations for the single-lane $I = 8$ and multi-lane $I = 22$ settings (5 independent runs $\times$ 5 episodes per run). Despite the substantial increase in traffic density and interaction complexity, MA-PMBRL exhibits only marginal utility degradation and maintains a stable distribution and maintains a stable distribution (low variance) in both utility and safety-related proxies, indicating robust generalization and favorable scalability to larger-scale multi-agent regimes.

*2) Impact of the Choice of $\phi$ in (8):* We evaluate the effect and robustness of our choice of $\phi$ in PGD by comparing it with the selection of the average solution $\phi_{\text{avg}}$ and the final solution $\phi_X$. As shown in Fig. 7, selecting the best solution ($\phi_{\text{best}}$) consistently yields the highest utility across episodes and ensures the fastest convergence to optimal utility, making it the most effective strategy. In contrast, the final solution ($\phi_X$) exhibits a slower increase in utility than the best solution and eventually stabilizes at a slightly lower performance level. This could be due to its potential sensitivity to abrupt state changes or insufficient convergence, especially in dynamic or unstable network conditions. Therefore, it may fail to capture the optimal policy, resulting in consistently higher variance. On the other hand, the average solution ($\phi_{\text{avg}}$) may overlook some local optima, especially under favorable vehicle dynamics or network conditions, as these optimal solutions can be "smoothed out" or diluted by the averaging process. As a result, although it gradually improves, it consistently underperforms both the best and final solution strategies. These findings indicate that consistently selecting the best solution outperforms other strategies, offering higher utility and faster convergence, making it the most effective choice for enhancing agent performance in the "Unprotected Intersection" scenario.

*3) Impact of Different Communication Ranges $d$:* We investigate the effect of the communication range $d$ on MA-PMBRL by varying $d$ from 0 to 200 (Fig. 8 and Table IV). As expected, Table IV shows that a larger $d$ substantially increases communication overhead, while Fig. 8 indicates improved learning efficiency, agility, and safety. However, beyond a threshold, further enlarging $d$ yields only marginal performance gains despite rapidly growing communication cost, revealing a clear performance–communication trade-off. To further examine this effect, we evaluate the average minimum clique-cover number $\bar{\chi}(\mathcal{G}_{d,k})$ versus $d$ over 20 independent trials. As shown in Fig. 9, $\bar{\chi}(\mathcal{G}_{d,k})$ decreases with $d$ and then saturates, consistent with both our theoretical analysis and the empirical trends in Fig. 8. These results underscore the need to balance communication efficiency and learning effectiveness in multi-agent coordination.

*4) Impact of Packet Loss:* To evaluate robustness under realistic CAVs communication impairments, we introduce dynamic packet loss to emulate intermittently connected environments (e.g., tunnels). We evaluate MA-PMBRL in the multi-lane setting with $I$=22 CAVs and communication range $d$=100 under loss

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TABLE V
COMPARISON OF THE MEANS AND STANDARD DEVIATIONS OF THE AVERAGE DECISION-MAKING TIME PER STEP FOR MARL METHODS EVALUATED ON THE SMARTS PLATFORM IN THE "FIGURE EIGHT" SCENARIO (UNIT: $1 \times 10^{-2}$ SECONDS)

| Scenarios | The Different MARL Methods | | | | |
|---|---|---|---|---|---|
| | MA-PMBRL | MBPO | FIRL | SVMIX | MA-SAC |
| **Single-lane loop** | $6.895 \pm 0.2931$ | $6.417 \pm 0.2664$ | $7.630 \pm 0.3549$ | $8.794 \pm 0.4038$ | $6.963 \pm 0.4457$ |
| **Multi-lane loop** | $8.742 \pm 0.3896$ | $7.906 \pm 0.3127$ | $9.779 \pm 0.3614$ | $10.689 \pm 0.4972$ | $8.407 \pm 0.5156$ |

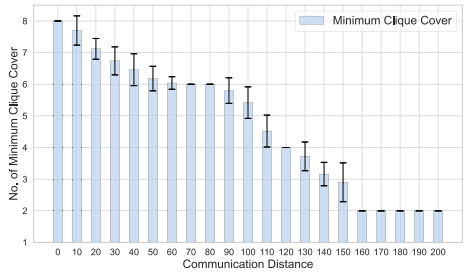



Fig. 9. Correlation between number of minimum clique cover $\bar{\chi}(\mathcal{G}_{d,k})$ and communication distance $d$.

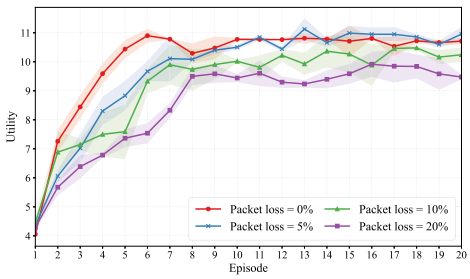



Fig. 10. Comparison of utility across different Packet Loss.

rates $\{0, 1\%, 5\%, 10\%, 20\%\}$, with five independent runs per setting; the results are summarized in Fig. 10. As shown in Fig. 10, increasing packet loss moderately slows convergence due to reduced availability of neighbor information; however, MA-PMBRL consistently attains a stable performance plateau and exhibits no instability even at a 20% loss rate. These results indicate that communication impairments mainly degrade sample efficiency rather than the reliability of the learned policy. Crucially, the pessimistic learning mechanism enables a graceful fallback to purely local decision-making when neighbor messages are missing, thereby preserving stability and safety under unreliable communication links.

*5) Impact of Rollout Length $L_{rollout}$:* Given the significance of rollout length to decision-making quality and overall utility performance, we experiment with different optimal rollout lengths, while presenting utility across the last five episodes after convergence. As shown in Fig. 11, the utility varies dramatically along with the variations in rollout length. Specifically, the shortest rollout length corresponds to lower utility and higher variability, suggesting that shorter rollouts lack sufficient information for accurate predictions. As the rollout length increases to 8 or 13, the utility improves with decreased variability, indicating that a moderate length allows for more informed, stable decisions.

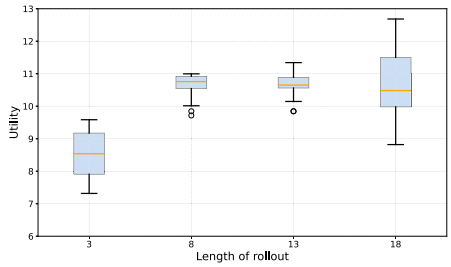



Fig. 11. Comparison of utility across different rollout lengths.

However, at a length of 18, the utility peaks but exhibits the most tremendous variability, indicating that longer planning horizons may improve optimization at the cost of stability. In conclusion, longer rollout lengths generally improve utility by accounting for more future states, but beyond a certain point, additional length offers diminishing returns. Considering that longer rollouts incur higher computational and storage costs, a length of 8 is defaultly chosen for the "Unprotected Intersection" scenario.

*6) Inference Time Analysis:* To assess the computational efficiency and real-time feasibility of MA-PMBRL, we report a runtime analysis in the "Figure Eight" scenarios. Table V summarizes the mean and standard deviation of the per-step decision-making time during testing after 20 training episodes, under both single-lane and multi-lane loop settings. Except MA-PETS, which relies on online planning, all compared methods, including MA-PMBRL, MAPPO, FIRL, SVMIX, and MA-SAC, perform direct policy inference, thereby enabling a fair comparison of inference latency. The results show that MA-PMBRL achieves inference time comparable to other policy-based baselines and substantially lower than the planning-based method, while exhibiting smaller variance and stable behavior across traffic settings. Notably, even in the multi-lane loop scenario, its per-step latency consistently meets the millisecond-level real-time requirement (e.g., a 100 ms control cycle).

## VII. CONCLUSION AND DISCUSSION

In this paper, we have studied decentralized MBRL for CAV self-decision control under communication constraints and partially covered offline datasets. We have proposed MA-PMBRL, which improves sample efficiency by enabling local sample sharing among neighboring vehicles within a communication range. Moreover, MA-PMBRL incorporates a pessimistic model-based min-max framework to enhance safety under model uncertainty and leverages PGD in the critic to efficiently derive control policies from pessimistic estimates. We have further established theoretical guarantees under communication constraints

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and partial coverage using graph-theoretic tools and concentration inequalities. Extensive experiments have validated that MA-PMBRL consistently outperforms representative MFRL and MBRL baselines and demonstrate the contribution of communication in multi-agent MBRL.

Several directions merit future study. Although MA-PMBRL converges rapidly and outperforms many baselines, a performance gap to the optimal baseline persists in some scenarios, largely due to the sensitivity of model-based methods to dataset quality. Nonetheless, its strong sample efficiency makes MBRL a promising pre-training paradigm to provide effective initialization for MFRL when data are scarce. Moreover, this work does not explicitly handle OOD data in MBRL, which may further limit performance. Finally, our theoretical guarantees are established for the SAC-based instantiation; extending the PAC guarantee to more general RL settings is an important direction.

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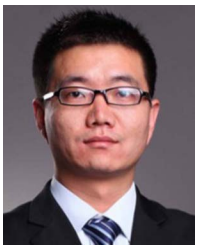
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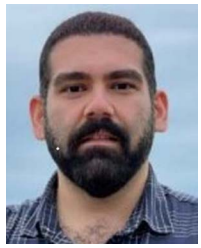
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## APPENDIX A
## TRAINING SPEED EVALUATION

TABLE I: Comparison of training speed among benchmark policies

| Category | Policy | Total Time/h |
|---|---|---|
| Heuristic | Random | 0.87 |
| | Round Robin | 0.90 |
| | Crash Avoid | 0.95 |
| DRL | DQN | 2.87 |
| | Rainbow | 3.23 |
| | REINFORCE | 2.33 |
| | PPO | 2.30 |
| | SAC | 3.41 |
| Ours | DDP-MORL | 8.52 |

## APPENDIX B
## DECISION LATENCY EVALUATION

TABLE II: Online Decision Latency Statistics Measured Periodically During Training

| **Epoch Interval** | **Mean (ms)** | **P95 (ms)** | **P99 (ms)** | **Min (ms)** | **Max (ms)** |
|---|---|---|---|---|---|
| 0 | 0.259418 | 0.384110 | 0.401058 | 0.228200 | 0.432600 |
| 10 | 0.283386 | 0.358065 | 0.432904 | 0.248000 | 0.516700 |
| 20 | 0.269302 | 0.317030 | 0.394784 | 0.242300 | 0.527300 |
| 30 | 0.308422 | 0.432425 | 0.497201 | 0.250700 | 0.507700 |
| 40 | 0.280927 | 0.334325 | 0.417887 | 0.246600 | 0.465200 |
| 50 | 0.310917 | 0.407850 | 0.591867 | 0.249200 | 0.635300 |
| 60 | 0.299990 | 0.376460 | 0.470604 | 0.247100 | 0.490400 |
| 70 | 0.329559 | 0.423650 | 0.541984 | 0.249400 | 0.702500 |
| 80 | 0.281974 | 0.340950 | 0.398761 | 0.247200 | 0.495300 |
| 90 | 0.288040 | 0.370375 | 0.472294 | 0.247200 | 0.564800 |
| 100 | 0.289046 | 0.332535 | 0.449425 | 0.243700 | 0.498300 |

## APPENDIX C
## VARIANCE & STABILITY ANALYSIS OF DRL AND HEURISTIC POLICIES

TABLE III: Comparison of Standard Deviation Across Different DRL Policies

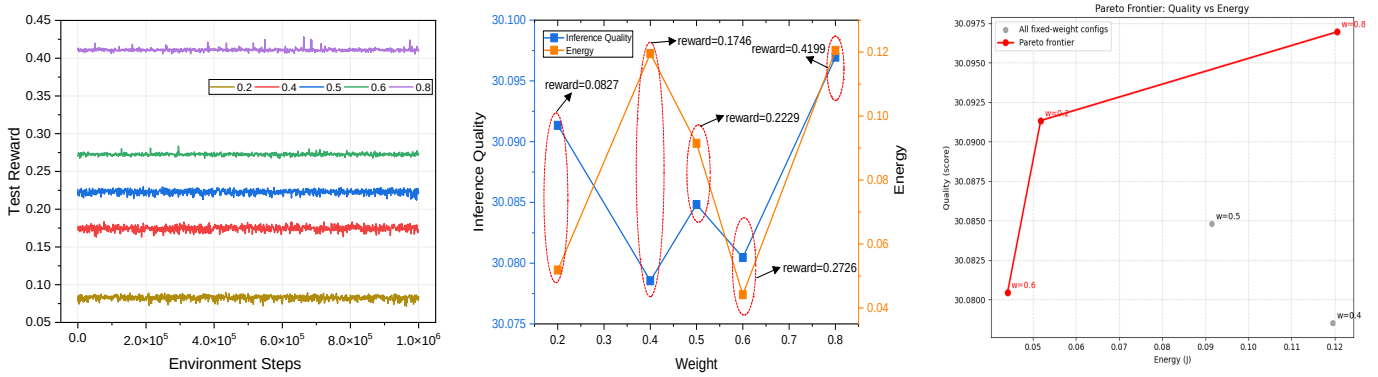
| **User Number** | **DDP-MORL Std** | **D2SAC Std** | **SAC Std** | **PPO Std** | **REINFORCE Std** | **DQN Std** | **Rainbow Std** |
|---|---|---|---|---|---|---|---|
| 5 | 9.1E−4 | 4.6E−3 | 6.2E−4 | 4.8E−4 | 4.7E−4 | 5.8E−4 | 7.1E−4 |
| 10 | 6.5E−4 | 4.4E−3 | 5.7E−4 | 5.6E−4 | 5.4E−4 | 5.2E−4 | 6.7E−4 |
| 15 | 6.2E−4 | 5.0E−3 | 4.9E−4 | 4.9E−4 | 4.9E−4 | 7.6E−4 | 8.5E−4 |
| 20 | 6.5E−4 | 4.4E−3 | 5.2E−4 | 6.1E−4 | 5.3E−4 | 0.00103 | 8.6E−4 |
| 25 | 6.2E−4 | 5.3E−3 | 5.6E−4 | 6.4E−4 | 5.6E−4 | 6.8E−4 | 5.7E−4 |
| 30 | 3.6E−4 | 4.6E−3 | 5.6E−4 | 4.8E−4 | 5.2E−4 | 5.5E−4 | 5.6E−4 |
| 35 | 4.0E−4 | 4.8E−3 | 5.6E−4 | 5.0E−4 | 4.9E−4 | 5.9E−4 | 6.1E−4 |

TABLE IV: Comparison of Standard Deviation Across Different Heuristic Policies

| User Number | DDP-MORL Std | Crash Avoid Std | Round Robin Std | Random Std |
|---|---|---|---|---|
| 5 | 9.1E−4 | 4.9E−4 | 8.9E−4 | 8.4E−4 |
| 10 | 6.5E−4 | 5.1E−4 | 6.2E−4 | 5.4E−4 |
| 15 | 6.2E−4 | 7.3E−4 | 0.00106 | 9.5E−4 |
| 20 | 6.5E−4 | 5.8E−4 | 9.4E−4 | 0.00108 |
| 25 | 6.2E−4 | 9.2E−4 | 8.6E−4 | 0.00186 |
| 30 | 3.6E−4 | 0.00143 | 0.00106 | 0.00136 |
| 35 | 4.0E−4 | 8.9E−4 | 5.5E−4 | 9.3E−4 |

## APPENDIX D
## ADDITIONAL EVALUATION OF THE WEIGHT ADAPTATION MECHANISM

**Discussion of Fig. 1.** Fig. 1 provides additional evidence on the effectiveness of the proposed weight adaptation mechanism. As shown in Fig. 1(a), different fixed weight settings lead to stable but significantly different test rewards, indicating that no single fixed weight can consistently achieve optimal performance. Fig. 1(b) further illustrates the trade-off between inference quality and energy consumption, where the achieved reward is highly sensitive to the manually selected weight. Finally, Fig. 1(c) presents the Pareto frontier between quality and energy, showing that adaptive weighting enables the policy to approach Pareto-optimal trade-offs and outperform most fixed-weight configurations. The corresponding quality–energy pairs are: ($w = 0.2$) (0.05184, 30.09134), ($w = 0.4$) (0.11954, 30.07855), ($w = 0.5$) (0.09149, 30.08483), ($w = 0.6$) (0.04414, 30.08046), and ($w = 0.8$) (0.12054, 30.09695). Under the objective of maximizing quality and minimizing energy, the configurations corresponding to ($w = 0.6$), ($w = 0.2$), and ($w = 0.8$) are non-dominated, whereas ($w = 0.4$) and ($w = 0.5$) are dominated.




(a) Test reward under different fixed weights.

(b) Quality–energy trade-off and reward vs. weight.

(c) Pareto frontier between quality and energy.

Fig. 1: Additional simulation results for validating the weight adaptation mechanism.